

HUMAN BIOLOGICAL SCIENCE SERIES

HBS

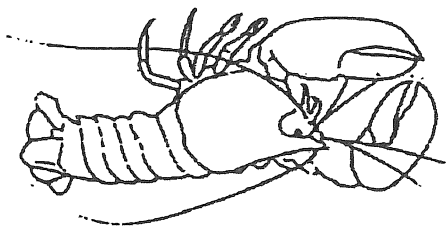
2A & 2B

Revision

Booklet

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MARRON PUBLICATIONS



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Science Dept
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The Practice of Human Biology Objectives

At the end of this unit students should be able to:

1. Define the term science.
2. Make observations using a range of senses.
3. Use scientific equipment to make more accurate observations.
4. Make inferences based on observations.
5. Explain the purpose of scientific method.
6. Write a hypothesis based on observations.
7. Define and identify independent, dependent and control variables.
8. Define and explain the purpose of experimental and control setups.
9. Conduct a simple experiment controlling a number of variables.
10. Identify the sample size in an experiment and explain the need for having a large sample size.
11. Explain the purpose and method of random selection for experimental groups.
12. Explain the meaning and influence of experimental bias.
13. Conduct a 'blind test'.
14. Explain why it is important to repeat experiments a number of times.
15. Discuss and identify biological variation and experimental error.
16. Identify the role of a placebo in the psychosomatic effect.
17. Record data accurately in an appropriate table.
18. Distinguish between continuous and discontinuous data.
19. Construct an appropriate graph for representing data.
20. Calculate the mean for a group of data.
21. Calculate percentage changes in data.
22. Write an appropriate conclusion for an experiment.
23. Design and conduct an extended investigation with a detailed method.
24. Identify weaknesses in experimental design and offer improvements.
25. Use scientific equipment appropriately.
26. Carry out experiments in a safe manner.
27. Work cooperatively with others.
28. Respect the opinions of others and carry out meaningful discussions.

6. Complete the following statement (on scientific method) using the words below.
(Experiment, observations, inferences, scientific method and hypothesis)

When investigating a question or problem, the scientist uses the (1) _____
_____ of phenomena thought to be
relevant to the problem; collation, classification and comparison of these observations
(i.e. interpretations or (3) _____); selecting one aspect of the problem and
formulating an appropriate (4) _____ to be tested; designing a controlled
(5) _____ to test the hypothesis.

7. Define the term Hypothesis.

8. Write a hypothesis for the following observations.
a) After completing a short run a student notices that their heart rate has increases.

b) After drinking a caffeinated drink a student notices that they have to urinate more.

c) After being out in the sun for one hour a student notices that their skin has become
darker.

9. Define the following variables.

a) Independent variable.

b) Dependent variable.

c) Controlled variable.

14. What is a placebo and how are they used to control the psychosomatic effect?

15. Data consists of measurements that are made during an investigation that are used to make determinations on the effect of the independent variable under consideration.
a) What is the difference between first and second hand data?

b) When and how should first hand data be recorded?

16. Explain the difference between continuous and discontinuous data.

17. Why is experimental data normally represented as a graph?

18. List five important characteristics of a line graph.

19. The following table represents second hand continuous data of the mass of a male human from birth to 26 years. Represent this data in a line graph on the following page.

Weight from birth to 26 years

Age(years)	Birth	2	4	6	8	10	12	14	16	18	20	22	24	26
Weight(kg)	3.5	12	16	22	27	32	38	46	58	63	67	70	72	73

a) Write the formula and describe how to calculate the mean (average) of a group of data.

20. Often to compare data sets percentage change is used as a measure of comparison. a) What is the formula for percentage change?

b) What was Craig's percentage increase in height over one year if his starting height was 170cm and his finishing height was 180cm?

c) Laura went on a diet over three months, what was her total percentage decrease in mass (over the three month period) if her starting mass was 69kg and her finishing mass was 65kg?

21. What types of information and statements should be included in the conclusion of an investigation?

22. A chemical company discovered a synthetic hormone, LACTPLUS, which appeared to make dairy cattle produce over 20% more milk. To test their idea they selected 50 dairy cattle and split them into two groups. To group A they administered LACTPLUS each day for a month and measured their milk production. At the same time they measured the milk production of group B but did not give them LACTPLUS. At the end of the month trial the average milk production of group A was 5.1 L/cow/day and for group B was 4.2L/cow/day.

a) Write the hypothesis they were testing.

b) What was the independent variable?

c) What was the dependent variable?

d) Give two variables that they would have controlled about the subjects (cows).

23. The school ball was coming up and some of the students decided they had to loose weight fast. A drug company was advertising a new drug, DISSOLVOFAT, which claimed it used as directed would drop 10% of your body mass in 10 days.

The students were a little skeptical so they went to the school principal (Mr. Svendsen) who directed them to the science department. Mr. Coultas in the science department claimed to be a weight loss guru and suggested students conduct a controlled experiment.

As it was not possible to conduct the test on students they chose the next best thing RATS! The experiment was conducted as follows. Fifty rats where randomly selected into two groups. Group A where given one treatment of DISSOLVOFAT with their food every morning. Group B was given a placebo.

Their initial and final average masses are recorded in the table below over the ten day period.

Average mass of Rats over a ten day period

Groups	Initial ave mass (g)	Final ave mass (g)	Percentage change
A	350.4	310.1	
B	352.6	345.5	

a) Write the hypothesis they were testing

b) What was the independent variable?

c) What was the dependent variable?

d) Give two variables that they would have controlled about the rats.

e) Give two variables other than about the rats that would also need to be controlled.

f) What was the experimental control?

g) What was the sample size?

The Practice of Human Biology Terminology

1. A positive statement that explains an observation:
2. First hand data collected using the body senses:
3. The practical application of scientific knowledge:
4. The variable being tested in an experiment:
5. Two ways that biological variation and experimental errors may be allowed for in experiments:
6. The main function of the control group:
7. Random selection is used to control this:
8. A possible explanation of an observation:
9. The variable that is measured in a controlled experiment to determine the effect of the independent variable:
10. The effect due to expectation:
11. A hypothesis may be proven by experimentation (T or F):
12. A major hypothesis that withstands the 'test of time':
13. A controlled experiment is designed to _____ a hypothesis:
14. Differences between individuals:
15. When asked to predict beyond the know data on a graph:
16. An inactive substance given to the control group:
17. Data where there is a relationship between the points:
18. Graph used to represent continuous data:
19. All variables that are kept constant in an experiment:
20. When asked to predict between two known values on a graph:

4. An organelle is a structure within the cell which is specialised for a particular function. Complete the following table for the functions of the following organelles.

Organelles	Function of Organelle
Cell Membrane	
Mitochondria	
Nucleus	
Lysosomes	
Ribosomes	
Endoplasmic Reticulum	
Cytoplasm	
Centrioles	
Golgi Body / Apparatus	
Nucleolus	

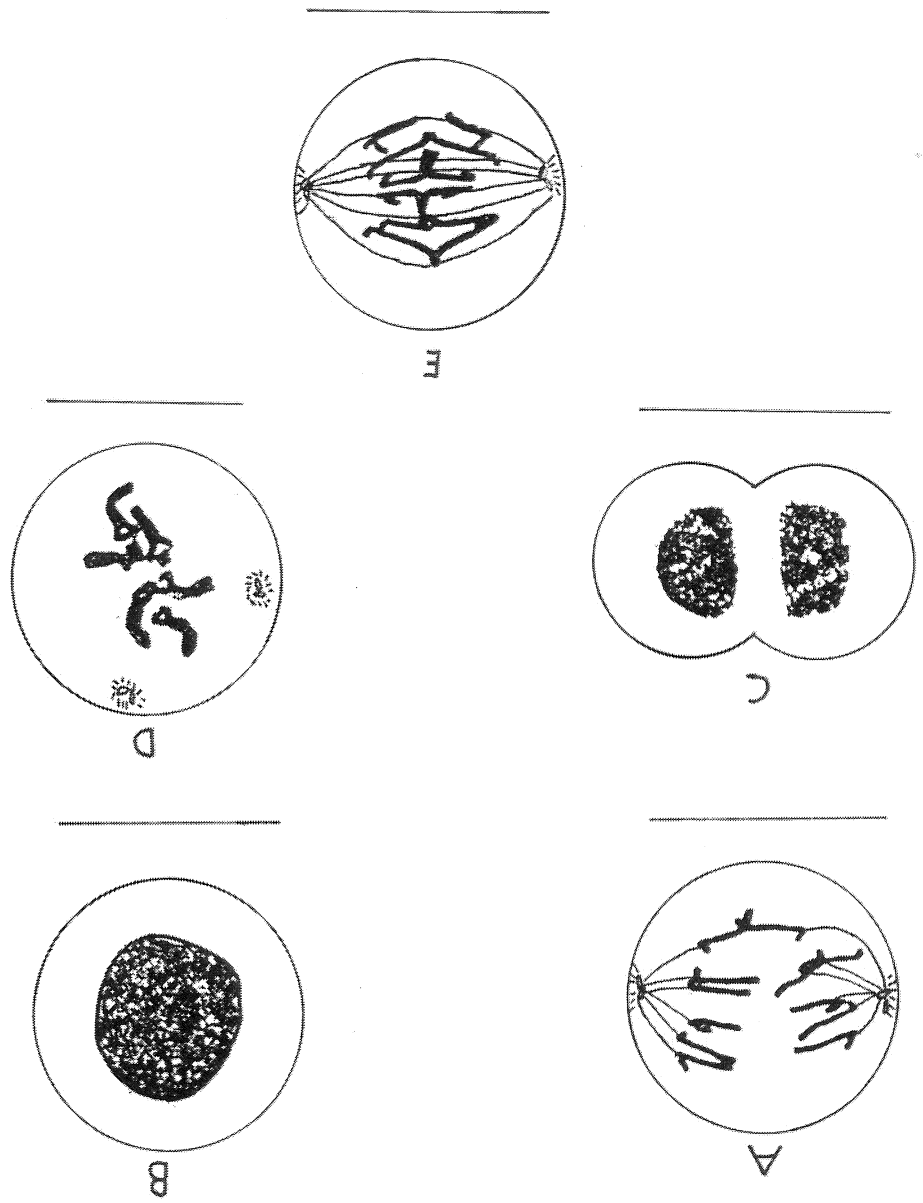
5. Mitosis is an essential process in the development of all organisms.

a) Define Mitosis.

b) What is the purpose of mitosis?

c) What is the difference between a Chromatid and a Chromosome?

h) Label each of the following cells with its appropriate stage of mitosis.



6. Explain what would happen to the number of chromosomes in body cells if there was no duplication of chromosomes before the cell underwent telophase.

Metabolism Objectives

At the end of this section students should be able to:

1. State the two main requirements, for the synthesis of cellular components.
2. Explain the following terms; synthesis, catabolism and anabolism.
3. Explain the following processes; cellular respiration, aerobic respiration and anaerobic respiration.
4. Write word equations for aerobic respiration and anaerobic respiration.
5. Explain where in the cell the following occur; aerobic respiration and anaerobic respiration.
6. Make comparisons between anaerobic and aerobic respiration.
7. Describe the ATP cycle.
8. Describe the nutrients required and their uses in metabolism.
9. Describe the structure of carbohydrates, proteins and lipids.
10. Define the following terms; catalyst and enzymes.
11. Explain the need for enzymes in living organisms.
12. List four properties of enzymes.
13. Draw diagrams to explain the lock and key model of enzyme action.
14. Describe the factors that can effect enzyme action.
15. Describe how enzymes become denatured.
16. Describe what is meant by the term 'optimal' conditions.

Metabolism Questions

1. What are the two main requirements for the synthesis of cellular components?
 - a) _____
 - b) _____

2. Explain using examples the following terms:

- a) Synthesis – _____
- b) Catabolism – _____
- c) Anabolism – _____

3. Define the following terms:

- a) Cellular respiration – _____
- b) Aerobic respiration – _____
- c) Anaerobic respiration – _____

7. A nutrient is a substance that is required for metabolism. There are six groups of nutrients; carbohydrates, proteins, lipids, vitamins, minerals and water. Complete the following table for the uses of nutrients in metabolism.

Nutrient	Use of nutrient in metabolism
Carbohydrate	
Protein	
Lipid	
Vitamins	
Minerals	
Water	

8. Describe the structure of the following.

a) Carbohydrates – _____

b) Proteins – _____

c) Lipids – _____

9. Chemical reactions are vital for cells to maintain their function
a) Define the following terms:

(i) Catalyst – _____

(ii) Enzyme – _____

b) Explain the need for enzymes in living organisms.

c) List four properties of enzymes:

(i) _____

(ii) _____

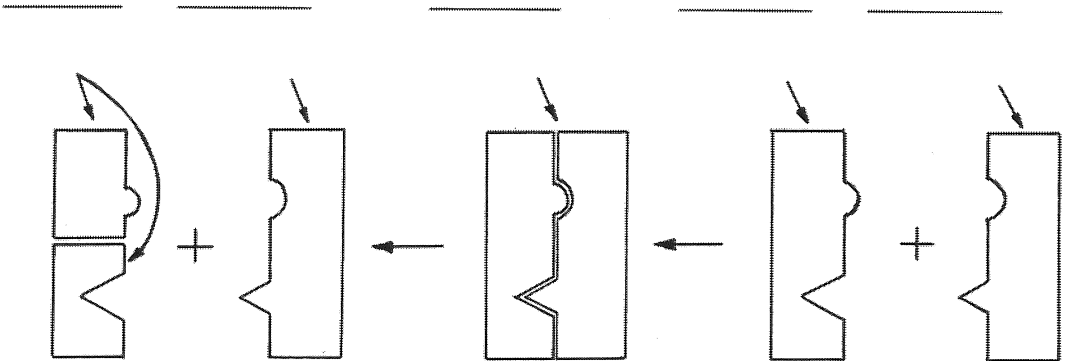
(iii) _____

(iv) _____

13. The action of enzymes is essential for chemical reactions to proceed within cells.

a) How are enzymes named?

b) How are enzymes thought to work?



c) Label the diagram below.

d) The diagram above shows what type of reaction?

e) Could the reaction in (c) above use the same enzyme in the opposite direction (ie synthesis)? Explain.

f) What term describes the specific shapes/places where the enzyme molecule and substrate 'fit' together?

g) If another chemical e.g. a poison happened to fit an active site, what would happen to the reaction?

h) What properties of enzymes are accounted for in the diagram in (c)?

Metabolism Terminology

1. The formation of new products from existing reactants:
2. The breaking down of large reactants into smaller products:
3. The formation of large products from small reactants:
4. Respiration requiring oxygen:
5. Respiration that results in the formation of lactic acid:
6. The location in a cell where the final stages of aerobic respiration takes place:
7. The acronym ATP stands for:
8. The name for the part of an enzyme that attaches to the substrate:
9. A substance required for metabolism:
10. Type of nutrient used for growth and repair:
11. Inorganic nutrient:
12. Type of nutrient that contains relatively small amounts of oxygen:
13. All enzymes are made of :
14. Enzymes are termed as biological or organic:
15. The name for the conditions where an enzyme works at its best rate:
16. This will happen to enzymes if they are placed in conditions outside of their 'range':
17. Enzymes work by reducing the _____ of the chemical reactions:
18. Vitamins act as _____ in enzyme controlled reactions:

2. The movement of materials in both directions across the cell membrane is necessary for a cell to function.
- a) Explain the difference between active and passive transport.

- b) Explain the difference between Endocytosis and Exocytosis.

- c) For the following processes state if they are active or passive.

(i) Diffusion – _____

(ii) Pinocytosis – _____

(iii) Active transport – _____

(iv) Phagocytosis – _____

- d) Define the following terms:

(i) Diffusion – _____

(ii) Facilitated diffusion – _____

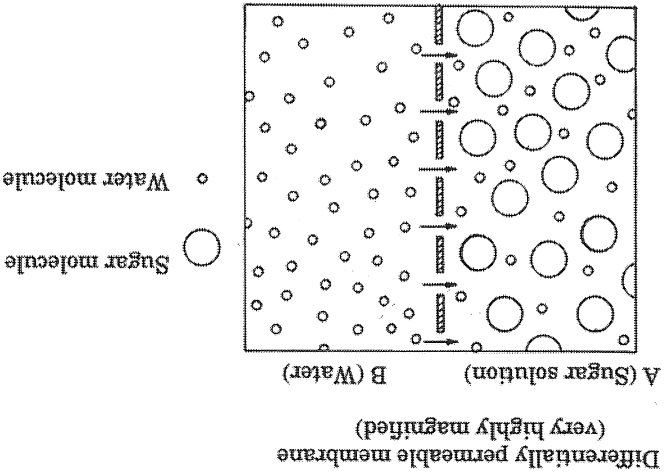
(iii) Osmosis – _____

(iv) Active transport – _____

(v) Pinocytosis – _____

(vi) Phagocytosis – _____

3. The diagram below shows a membrane separating a sugar solution from distilled water.



a) In which direction will the sugar diffuse? Explain.

b) In which direction will water diffuse?

c) In which direction will there be a net diffusion of water? Explain.

d) What will happen to the osmotic pressure in side A?

e) If the membrane was permeable to both sugar and water:
(i) Which molecule would diffuse faster? Explain.

(ii) After some time, what would you notice about the concentration of the sugar solutions in sides A and B?

(iii) At equilibrium, would diffusion cease? Explain.

Circulatory System Objectives

At the end of this unit students should be able to:

1. Describe the main functions of the circulatory system.
2. Describe the three main components of the circulatory system.
3. Describe why multi-cellular organisms require a circulatory system.
4. Label a diagram of the heart.
5. Describe the functions of the various parts of the heart.
6. Explain the cardiac cycle.
7. Describe cardiac output.
8. Explain the difference between systemic and pulmonary circulation.
9. Describe the structure and function of arteries, veins and capillaries.
10. Compare the structure and function of arteries and veins.
11. List the four major components of blood.
12. Describe a number of important functions of blood.
13. Describe the process of blood clotting.
14. Describe the inflammation response.
15. Complete a table on the origin, function, lifespan and fate of the different blood cells.
16. Describe the different types of white blood cells.
17. Describe the role of plasma in the blood.
18. Explain how oxygen and carbon dioxide are transported in the blood.
19. Describe how blood flow is regulated.

6. For each of the parts labelled in questions 5 write its name and briefly describe its function;

a) _____ - _____

b) _____ - _____

c) _____ - _____

d) _____ - _____

e) _____ - _____

f) _____ - _____

g) _____ - _____

h) _____ - _____

i) _____ - _____

j) _____ - _____

k) _____ - _____

l) _____ - _____

m) _____ - _____

n) _____ - _____

o) _____ - _____

p) _____ - _____

q) _____ - _____

7. Use arrows to show the direction of blood flow through the heart in question 5 and shade the heart red where there is oxygenated blood and blue where there is deoxygenated blood.

12. Briefly describe the function of each of these circulatory systems;

a) Systemic – _____

b) Pulmonary – _____

c) Portal – _____

13. What side of the heart is involved with systemic circulation?

14. What side of the heart is involved with pulmonary circulation?

15. What are the advantages of the double circulation systems of humans have over a single circulation system?

16. Complete the following table comparing the structure of arteries, veins and capillaries.

Blood vessel	Relative thickness of wall	Valves present/absent	Muscle present	Size of lumen
Arteries				
Veins				
Capillaries				

17. Describe the function of the following blood vessels;

a) Arteries – _____

b) Veins – _____

c) Capillaries – _____

25. List the four major components of blood.

- a) (Non-cellular) – _____
- b) (Cellular) – _____
- c) (Cellular) – _____
- d) (Cellular) – _____

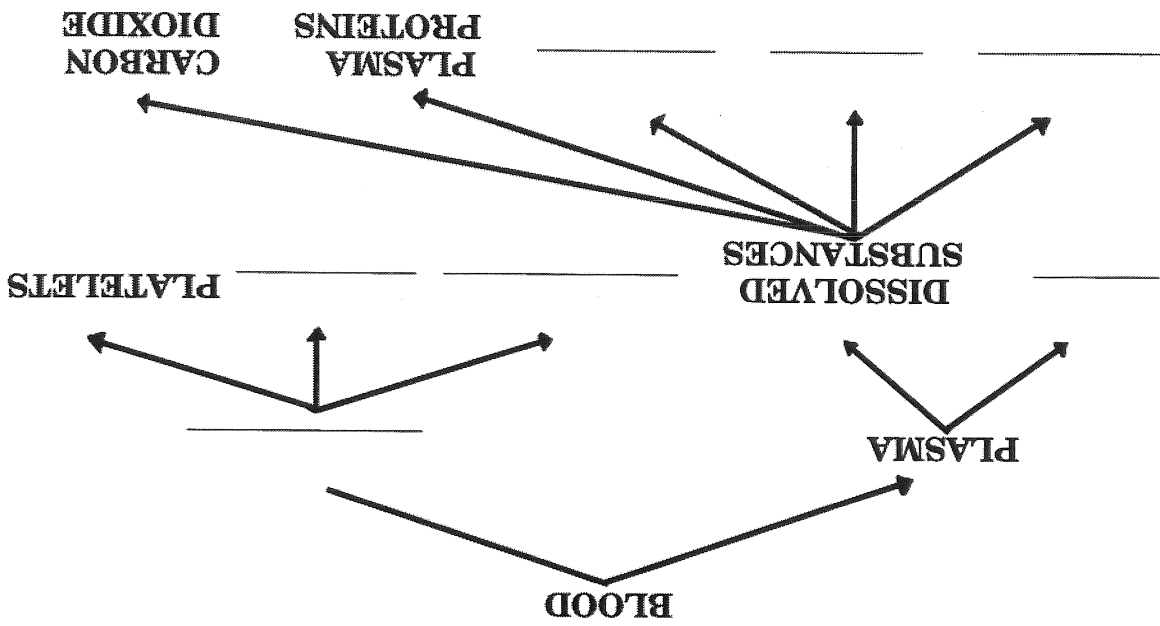
26. Briefly describe seven main functions of blood.

- a) _____
- b) _____
- c) _____
- d) _____
- e) _____
- f) _____
- g) _____

27. List five substances that are transported in the plasma.

- a) _____
- b) _____
- c) _____
- d) _____
- e) _____

28. Complete the diagram below on the composition of blood using the following words.
(Nutrients, erythrocytes, blood cells, water, leucocytes, hormones, wastes)



35. Define the following terms.

a) Plasma – _____

b) Tissue (intercellular) fluid – _____

c) Cytoplasm (intracellular fluid) – _____

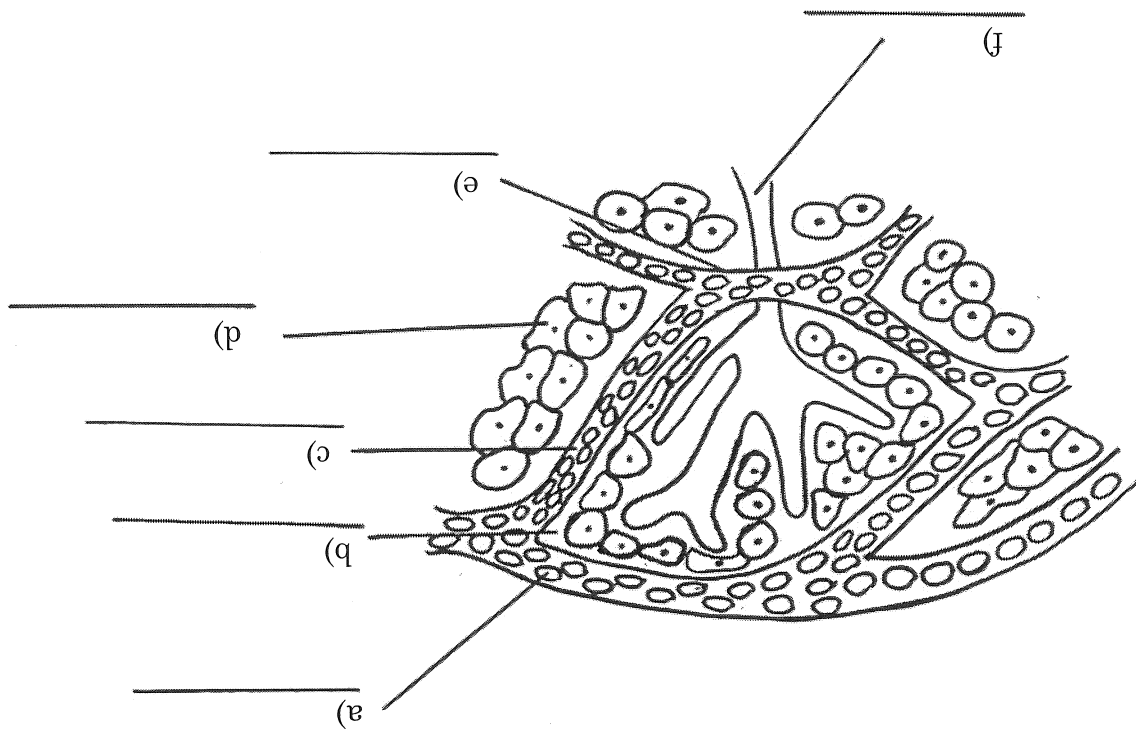
d) Lymph – _____

e) Diffusion – _____

36. List the five substances that are exchanged between capillaries, tissue fluid and cytoplasm.

- a) _____
- b) _____
- c) _____
- d) _____
- e) _____

37. Label the following diagram of the relationships between various body fluids.



Circulatory System terminology

1. The colour typically used on diagrams to represent oxygenated blood:

2. The upper chambers of the heart:

3. The lower chambers of the heart:

4. The vessels that carry blood from the heart to the lungs:

5. The type of blood in pulmonary veins:

6. The type of valves that prevent backflow of blood from the ventricles to the atria:

7. The type of valves that prevent backflow of blood from the arteries to the ventricles:

8. The type of circulation that transports blood to and from the lungs:

9. The ventricle with the thickest walls:

10. The side of the heart involved in systemic circulation:

11. The arteries that carry blood to the intestines:

12. The arteries that carry blood to the head:

13. The arteries that carry blood to the heart muscle:

14. The type of blood vessel with the thickest walls:

15. The type of blood vessel with the largest lumen:

16. They type of muscle within the walls of arterioles:

17. The name for the small veins that join to capillaries:

18. The liquid component of blood:

19. The scientific name for red blood cells:

20. The scientific name for white blood cells:

21. The scientific name for platelets:

22. The shape of red blood cells is described as:

23. Erythrocytes cells are formed in the:

24. The organ responsible for removing old and dead blood cells from the blood:

25. The term for when the heart is relaxing:

26. The instrument used to measure blood pressure:

27. The term for pressure in the arteries during ventricular Systole (contraction):

Respiratory System Objectives

At the end of this section students should be able to:

1. Explain the need for a gas exchange system in multi-cellular organisms.
2. Label a diagram of the respiratory system.
3. Describe the structure and function of the parts of the respiratory system.
4. Define the following terms; breathing, inspiration, expiration and cellular respiration.
5. Describe the direction of air flow from the nose/mouth to the alveoli.
6. Describe the mechanics of inspiration and expiration.
7. Understand that it is the concentration of carbon dioxide in the blood that controls breathing rate.
8. Describe the following term; diffusion, diffusion gradient and gas exchange.
9. Describe what a respiratory surface is and list four properties of a respiratory surface which aid diffusion.
10. Explain the process and mechanisms involved in the diffusion of gases between air in the alveolus and the blood.
11. Outline the differences in composition between inhaled and exhaled air.
12. Explain how the following mechanisms are involved in keeping the lungs clean; coughing, mucous secretion and cilia.
13. Explain the cause, symptoms and treatment of the following; asthma, emphysema, lung cancer, bronchitis, carbon monoxide poisoning and Pneumonia.

4. Explain the structure and function of the following.

Structures	Structure/function
Tracheal Cartilage	
Trachea	
Intercostal muscle	
Rib	
Left bronchus	
Pleural Cavity	
Alveoli	
Bronchiole	
Pleural membranes	
Diaphragm	
Nasal cavity	
Epiglottis	
Pharynx	
Larynx	
Lung	

5. Define the following terms;

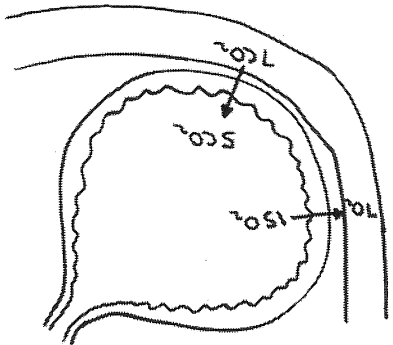
a) Breathing – _____

b) Inspiration – _____

c) Expiration – _____

d) Cellular respiration – _____

12. Using the diagram below showing the concentration of oxygen and carbon dioxide in the alveolar capillaries and air. Explain why carbon dioxide moves from the blood in the alveolar capillaries into the alveolar air, and why the oxygen moves from alveolar air into the alveolar capillaries.



13. Name and describe four mechanisms that maintain the diffusion gradient in the lungs.

14. Complete the following table contrasting the composition of inhaled air and exhaled air.

Component	Inhaled Air	Exhaled Air
Oxygen (%)		
Carbon Dioxide (%)		
Water Vapour (%)		

Respiratory System Terminology

- 1. System involved in the exchange of gases:
- 2. The movement of air into and out of the lungs:
- 3. Filters, warms and moistens the air:
- 4. Fine hairlike projections that filter the air:
- 5. Smell receptors:
- 6. Produced by mucus membrane and help to trap dust particles:
- 7. Common pathway for air and food:
- 8. Organ of speech:
- 9. Two folds of mucus membrane that vibrate to produce sound:
- 10. Can be increased by increasing the force air over the vocal chords:
- 11. Changing the tension of the vocal chords changes this:
- 12. Stops food going down the trachea:
- 13. Largest tube in system contains rings of cartilage:
- 14. There are two of these one to each lung:
- 15. Tubes that have non cartilage but smooth muscle:
- 16. Breathing in :
- 17. Breathing out:
- 18. Double membrane which helps to reduce friction:
- 19. Muscles used to raise ribs:
- 20. Dome-shaped muscle that separates thoracic and abdominal cavities:
- 21. Small 'air sacs' where gas exchange takes place:
- 22. Made of bone and help to protect the lungs:
- 23. Gas which controls breathing rate:
- 24. Oxygen and haemoglobin combine to produce this:
- 25. Most carbon dioxide is carried in the plasma as this:
- 26. Combination of carbon dioxide and haemoglobin:
- 27. Oxyhaemoglobin is this colour:
- 28. Type of cancer for which smoking is a major cause:
- 29. Disease where the alveoli loose their elasticity:

Digestive System Objectives

At the end of this section students should be able to:

1. Define the following terms ingestion, digestion and defecation (egestion).
2. Distinguish between mechanical and chemical digestion.
3. Label a diagram of and describe the uses for the four types of teeth.
4. Describe how swallowing takes place.
5. Describe what is peristalsis and where does it take place in the body.
6. Describe the substrates acted on and the products produced for the following enzymes; amylase, lipase and protease.
7. Complete a table for the main enzymes involved in chemical digestion in the body.
8. Describe a number of tests for starches, proteins, glucose and lipids.
9. Describe the role of bile and HCl in digestion.
10. Label a diagram of the human digestive system.
11. State the functions of the following parts of the human digestive system; mouth, salivary gland, pharynx, oesophagus, epiglottis, teeth, tongue, stomach, duodenum, jejunum, ileum, colon, appendix, rectum, anus, gall bladder, pancreas, liver and caecum.
12. Label a cross section of the alimentary canal.
13. Describe the features of the small intestine (ileum) that make it suitable for the absorption of digested nutrients.
14. Describe how the surface area of the ileum is increased.
15. Explain how absorption of nutrients takes place out of the ileum into the bloodstream.
16. Explain the concept of a balanced diet.
17. Explain the problems associated with the following diets; high salt, low fibre, high sugar, high saturated fats and excess kilojoules.
18. Explain the causes symptoms and treatments of the digestive disorders; ulcers, indigestion, constipation and bowel cancer.

4. Complete the following table.

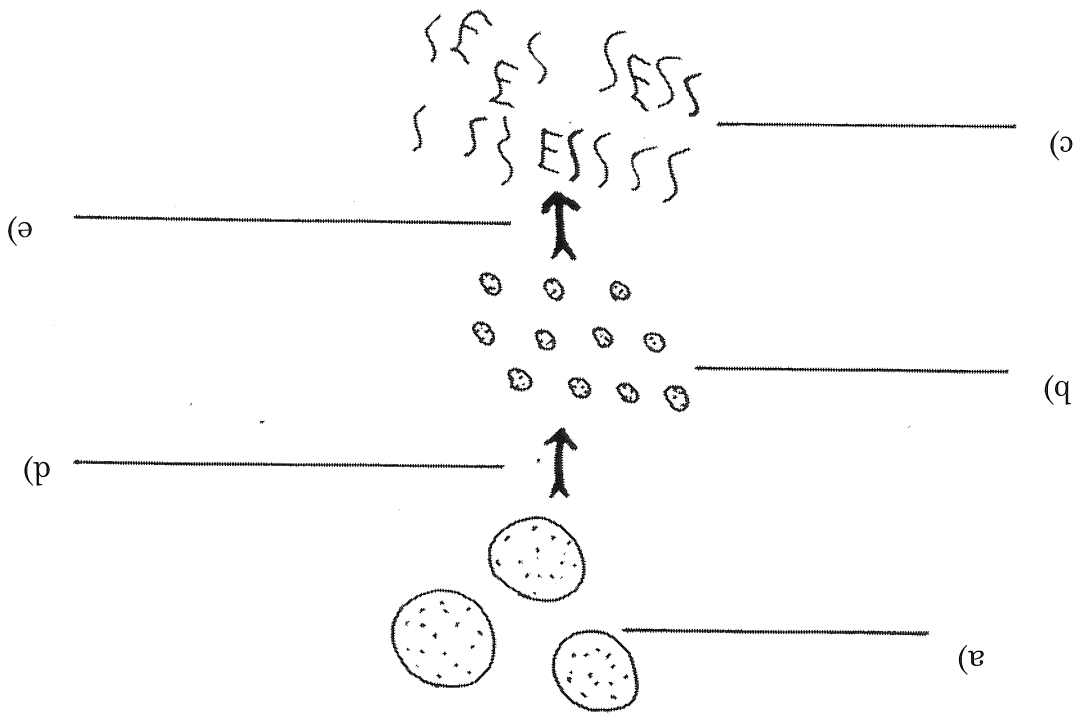
Tooth type	Total number	Description	Function
Incisors			
Canines			
Premolars			
Molars			

5. Outline the process of chemical digestion including the general parts of the digestive system involved.

6. Discuss with your teacher which parts of the following table are required knowledge.

Enzyme	Source	Site of action	Substrate	Products
Salivary amylase				
Pancreatic amylase				
Maltase				
Sucrase				
Lactase				
Lipase				
Gastric protease				
Pancreatic protease				
Peptidase				

9. Label the following diagram of chemical digestion lipids.



10. Complete the following table outlining tests that can be used to test for the presence of specific substances in foods.

Substance	Test	Colour change for positive result
Starch		
	Benedict's	
Protein		
	Brown paper	

11. Describe the role of each of the following in chemical digestion.

- a) Bile – _____
- b) Pepsinogen – _____
- c) Trypsinogen – _____
- d) Gastric HCL – _____

14. For each of the following parts labelled on the diagram on the previous page briefly outline its function(s).

a)	
b)	
c)	
d)	
e)	
f)	
g)	
h)	
i)	
j)	
k)	
l)	
m)	
n)	
o)	
p)	
q)	
r)	
s)	

19. Define the following terms and describe their role in the digestive system.

a) Secretion –

b) Absorption –

c) Peristalsis –

d) Defecation –

20. Explain one problem associated with each of the following diets.

a) High salt –

b) Low fibre –

c) High sugar –

d) High saturated fat –

e) Excess kilojoules –

8. Organ that contains gastric juice:
9. Teeth used for cutting or biting:
10. Used for crushing and grinding food:
11. Produces bile:
12. Stores bile:
13. First part of the small intestine mainly involved in chemical digestion:
14. The taking in of food and water:
15. Back of the mouth cavity:
16. Carries food from the mouth to the stomach:
17. Circular muscle that control the flow of material from the stomach to the duodenum:
18. Final part of the large intestine where faeces is stored:
19. Enzyme in the stomach helps to breakdown proteins to polypeptides:
20. Organic catalyst which speeds up chemical reactions without itself being consumed:
21. Lining of the stomach:
22. Helps to break-up fat into smaller particles:
23. Finger-like projections that extend out of the mucosa in the small intestine:
24. Movement of materials against the concentration gradient and requires metabolic energy:
25. Enzyme involved in the breakdown of lipids:
26. Intestine which has the biggest diameter:
27. The removal of faeces out of the rectum:
28. Can be caused by a lack of roughage:
29. Frequent defecation of watery faeces:
30. Common form of cancer:
31. Has no specific function:
32. Produces pancreatic juice containing enzymes:
33. Opening surrounded by the anal sphincter:
34. Amylases tend to break these down:
35. Mainly removed in the large intestine:

Excretory System & Assisted Protection of the Body Questions

1. Define the following terms.

a) Internal environment –

b) Interacellular fluid (cytoplasm) –

c) Intracellular fluid –

d) Extracellular fluid –

f) Plasma –

g) Regulation –

2. Why is it important to maintain constant levels of pH, nutrients and wastes to maintain efficient function of cells?

a) pH –

b) Nutrients –

c) Wastes –

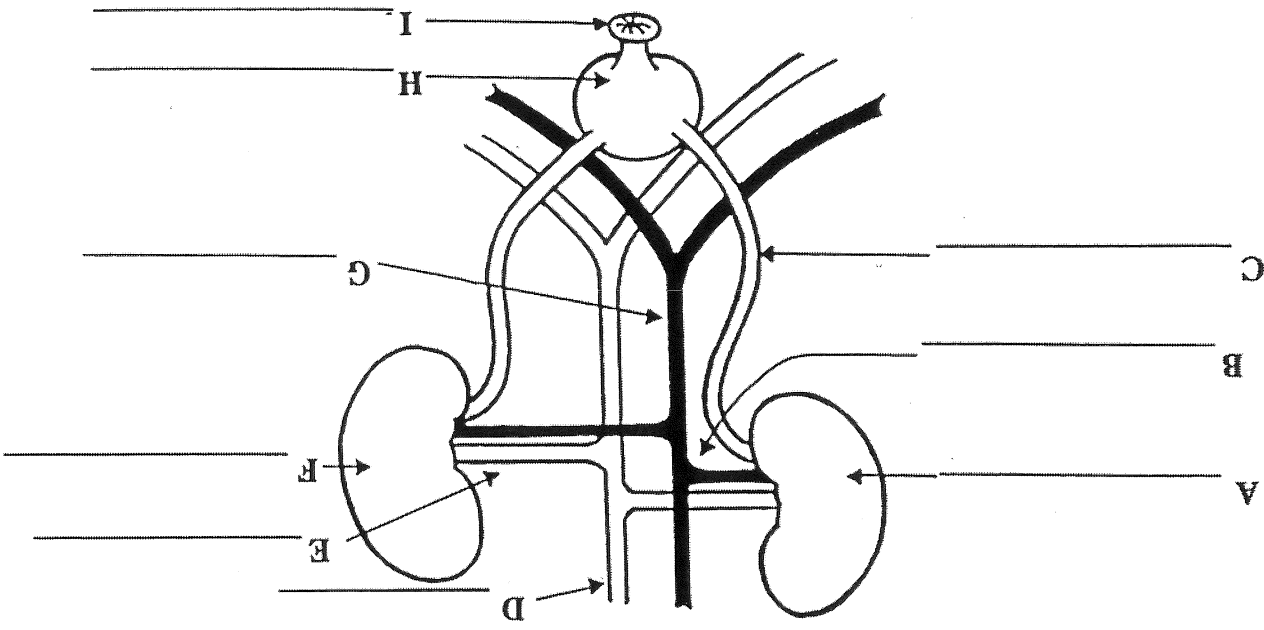
3. The concentration of substances in the cytoplasm, intercellular fluid and plasma differs according to the activities taking place. Use the words below to complete the following passage.
(Carbon Dioxide, Fluids, Permeability, Concentration, Materials, Proteins, Cytoplasm, Membrane)

“There is a constant exchange of (1) _____ between the plasma, intercellular fluid and intracellular (2) _____. A general rule for small molecules is the (3) _____ of substances in the plasma directly affects concentrations in the intercellular fluid and in turn the (4) _____ eg nutrients diffuse from the plasma through the intercellular fluid to the cytoplasm. The opposite occurs for small molecules moving from the cytoplasm to the plasma eg (5) _____. Larger molecules like (6) _____ are limited in their movement due to the (7) _____ of the walls of the blood vessels and the cell (8) _____.”

8. List the four functions of the kidney.

- a) _____
- b) _____
- c) _____
- d) _____

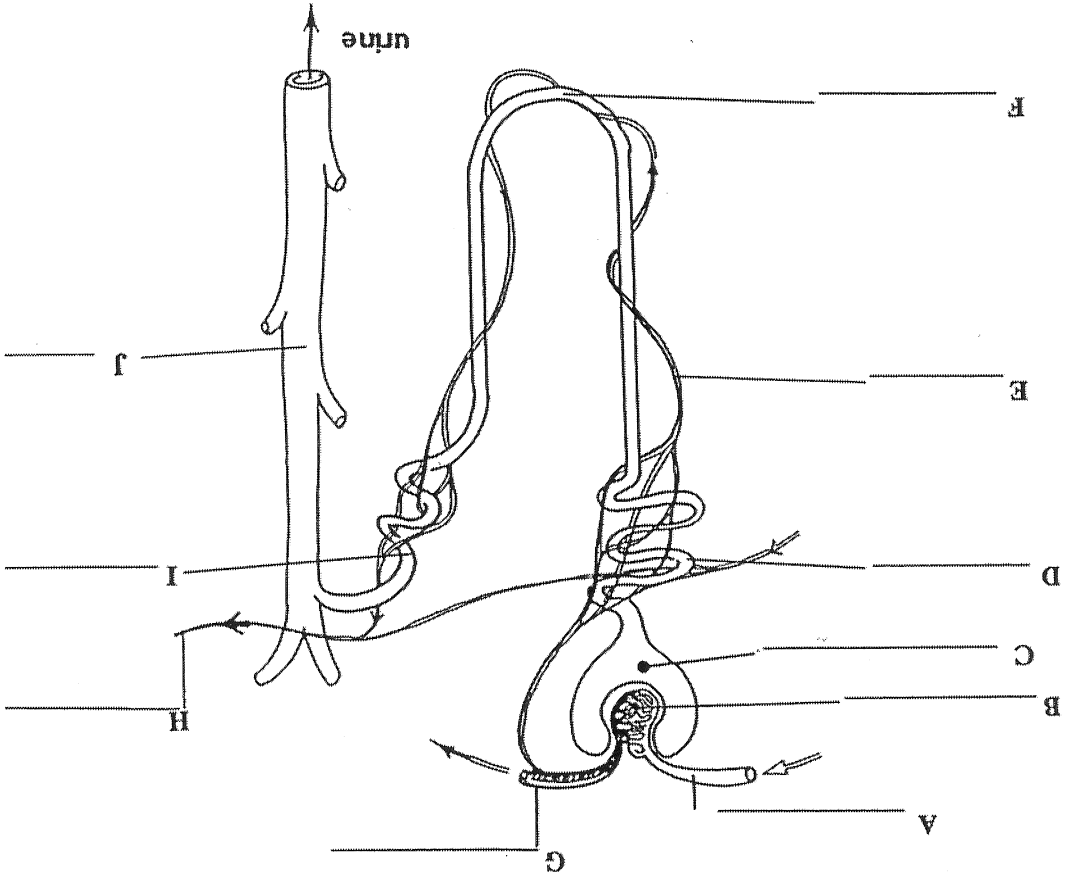
9. Label the following diagram of the Urinary System.



10. Give the functions of the parts of the Urinary System?

- a) Renal artery – _____
- b) Renal vein – _____
- c) Ureter – _____
- d) Urethra – _____
- e) Bladder – _____

13. Each kidney is made up of around 1 million nephrons, these are functional units of the kidney. Label the following diagram of the Nephron.



14. What is the function of each of the following parts of the nephron?

a) Glomerulus – _____

b) Glomerular Capsule – _____

c) Proximal Convoluted tubule – _____

d) Loop of Henle – _____

e) Distal Convoluted Tubule – _____

f) Collecting Duct – _____

g) Afferent Arteriole – _____

h) Efferent Arteriole – _____

i) Capillary Network – _____

Excretory System & Assisted Protection of the Body Terminology

- 1. Fluid between the cells:
- 2. Fluid inside the cell:
- 3. Liquid part of the blood:
- 4. Toxic substances that build up in the blood:
- 5. Substances required by cells:
- 6. A membrane that allows some substances to pass through but not others:
- 7. Breakdown of proteins in the liver:
- 8. Carries blood to the kidney:
- 9. Carries blood away from the kidney:
- 10. Stores urine:
- 11. Non toxic nitrogenous waste:
- 12. Carries urine to the bladder:
- 13. Takes the urine from the bladder to outside the body:
- 14. Tough, waterproof, protective barrier:
- 15. Sticky and helps to trap foreign particles:
- 16. Ball of blood vessels in nephron:
- 17. Loop of:
- 18. Duct the helps to control the amount of water absorbed back into the blood:
- 19. Functional unit of the kidney:
- 20. Removal of faeces from the body:

4. Use the words below to complete the following summary on the structure and function of DNA. *Note – The numbers refer to the answers.*
- (bases, ribosome's, double, structure, replicate, strand, helix, metabolism, identical, sequence, synthesis, proteins, paired, complementary, pairs, RNA, phosphate, function.)

A molecule of DNA consists of a (1) _____ chain of alternating sugar and (2) _____ held together by (3) _____ nitrogen bases. This structure is then twisted to form a double (4) _____.

The genetic information is stored in the (5) _____ of the nitrogenous (6) _____ bases only form (7) _____ pairs, so the code on one side of the (8) _____ is the equivalent of the other side.

The complementary base pairs allow the DNA molecule to (9) _____ itself. When DNA "unzips" (splits in the middle) each side can rebuild itself using the complementary base (10) _____ therefore, producing two (11) _____ strands.

The DNA is able to release information from the nucleus by splitting, copying itself in the form of (12) _____, which then passes out of the nucleus to the (13) _____ where it determines protein (14) _____ in the cell.

The nucleus is able to control the cell by determining what (15) _____ the cell can make. It controls the (16) _____ of the cell as most cell structures are protein based. It controls the (17) _____ of the cell as all enzymes are proteins and enzymes determine cell (18) _____.

DNA Terminology

1. What does the acronym DNA represent: _____
2. The identical structures that form a chromosome: _____
3. The point of attachment of chromatids in a chromosome: _____
4. The smallest inheritable unit of DNA: _____
5. DNA in a non dividing cell: _____

Inheritance, Meiosis & Variation Questions

1. What is genetics and describe two reasons why the study of human genetics is important.

important.

2. The passing of characteristics from parents to offspring is termed heredity. Offspring produced by asexual reproduction are usually very similar to their parents. However offspring produced by sexual reproduction tend to show variation.

offspring produced by sexual reproduction tend to show variation.

- a) What is meant by the term variation in humans?

- b) Explain why variation occurs more often in sexually reproducing populations.

- c) List five ways in which humans vary.

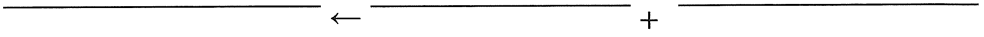
3. Describe what is meant by the following terms.

- a) Genotype –

- b) Phenotype –

- c) Environment –

- d) Complete the following using the terms above;



6. The structure of chromosomes allows the DNA to be moved about the nucleus/cell during cell division.

a) The human cell contains _____ chromosomes or _____ pairs of chromosomes.

b) The human cell contains two types of chromosomes autosomal and sex chromosomes. Describe the function of each and state the number of each type in the cell.

i) Autosomal – _____

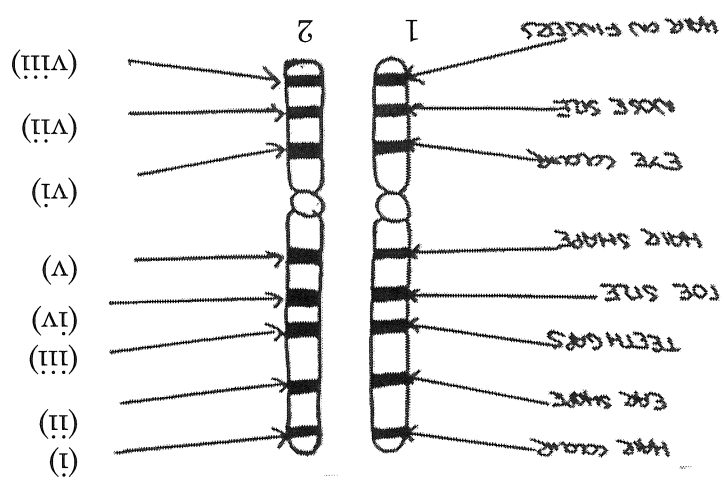
Chromosomes consisting of _____ pairs

ii) Sex – _____

Chromosomes consisting of _____ pair

c) Describe a homologous chromosome.

d) Label the diagram of a pair of homologous chromosomes to indicate where the other gene of the pair would be located.



e) Explain why the two chromosomes in (d) above are described as homologous.

f) In the diagram (d) above explain how the chromatids (1) and (2) are; (i) Structurally the same.

(ii) Possibly functionally different.

10. Many characteristics of organisms are controlled by a pair of alleles with one allele of the pair coming from the mother and the other from the father.

a) What is the name given to alternative forms of a gene?

b) What is monogenic inheritance?

c) What type of phenotype variations are produced for a monogenic characteristic?

d) List four traits which are controlled by monogenic inheritance.

11. Explain the difference between dominant and recessive alleles.

12. Explain the difference between homozygous and heterozygous genotypes.

13. In dominant/ recessive inheritance the dominant allele is normally assigned the _____ letter and the recessive _____ letter.

14. Human genetics is very complex; in this course you have studied a number of simplified examples to help understand the basics of genetic inheritance.

a) In Humans, the allele for brown eyes masks (is dominant over) the effect of the allele for blue eyes. Use the letters given below to complete the Genotypes for the following Phenotypes (Brown = B, Blue = b)

i) Homozygous brown – _____

ii) Homozygous blue – _____

iii) Heterozygous brown – _____

b) Why is it not possible to have a heterozygous blue eyed person?

e) A homozygous blue eyed male and a heterozygous brown eyed female have a large family of eight children. Two have blue eyes and six have brown eyes. According to genetic probability (i.e. part c) you would have expected four blue and four brown eyed children. Explain how this discrepancy could take place.

16. Free ear lobes are dominant to attached ear lobes in humans. A man with a heterozygous genotype has children with a woman who has attached ear lobes. What are the probable genotypic and phenotypic frequencies for the ear lobes of their children? Show your working.

Genotypes – _____

Phenotypes – _____

17. Astigmatism is dominant to normal vision in humans. A woman with astigmatism, whose father had normal vision, has children to a man with normal vision. What is the probability that they will have a child with normal vision? Explain your reasoning and show your working.

18. Roman nose (R) is dominant to straight nose (r). A roman nosed man whose mother had a straight nose has children to a straight nosed woman. They have five children, three with Roman nose and two with straight nose.

List the genotypes of all the individuals above.

- a) Roman nosed man – _____
- b) Straight nosed woman (mans mother) – _____
- c) Straight nosed woman – _____
- d) Straight nosed children – _____
- e) Roman nosed children – _____

(ii) Heterozygous X Heterozygous.

c) The genotype Hb^S Hb^S causes the individual to die but genotype Hb^A Hb^S has some advantages and disadvantages. Describe some of these.

- Advantage (i) –
- Disadvantages (i) –
- (ii) –

22. Why are males more likely to have sex-linked abnormalities than females?

23. Characteristics determined by alleles on the sex chromosomes are termed sex-linked. a) Describe sex linked inheritance.

b) Are the majority of the sex-linked abnormal characteristics carried on the sex (X) chromosome, sex-linked dominant or sex-linked recessive?

c) Red-Green colour blindness is an example of a sex-linked recessive trait. Using the following information complete the table below. X^N = Normal sight, Xⁿ = Colour Blind

Phenotype	Genotype
Colour blind male	
Normal male	
Carrier female	
Normal female	
Colour blind male	
Colour blind male	

Inheritance, Meiosis & Variation Terminology

1. The method of inheritance of red-green colour blindness:
2. The term for the alternative form of a gene:
3. The term for the passing of characteristics from parents to offspring:
4. The term for the actual gene complement of an organism:
5. The term for the way a genotype is expressed:
6. The type of twins produced from a single fertilized ovum:
7. The term for non-sex chromosomes:
8. The term for structurally identical chromosomes:
9. The type of cell division for gamete production:
10. The term for the number of chromosomes in a somatic cell:
11. The term for the fusion of sperm and ovum:
12. The term for a characteristic determined by a single gene:
13. The type of genotype when the two alleles are different:
14. Term describing characteristics located on the X chromosome:
15. The genotype of a female carrier:
16. The genetic condition when blood does not clot:
17. The term for alleles that must be homozygous to be expressed:
18. The term for when chromosomes or chromatids do not correctly separate:
19. The parent that determines the sex of an individual:
20. The standard symbol used for a dominant allele:
21. Rod-like structure made of genes that appears in the nucleus before cell division:
22. Organelle in the cytoplasm that manufactures proteins:
23. Unit from which proteins are manufactured:
24. Differences between individuals:
25. Reproduction that involves the joining of gametes:
26. Term used to describe half the normal number of body chromosomes:
27. Type of cell division for growth and repair:
28. When both alleles are expressed in the heterozygote:

Pedigrees, Genetic Testing and the Human Genome

Project Objectives

At the end of this unit students should be able to:

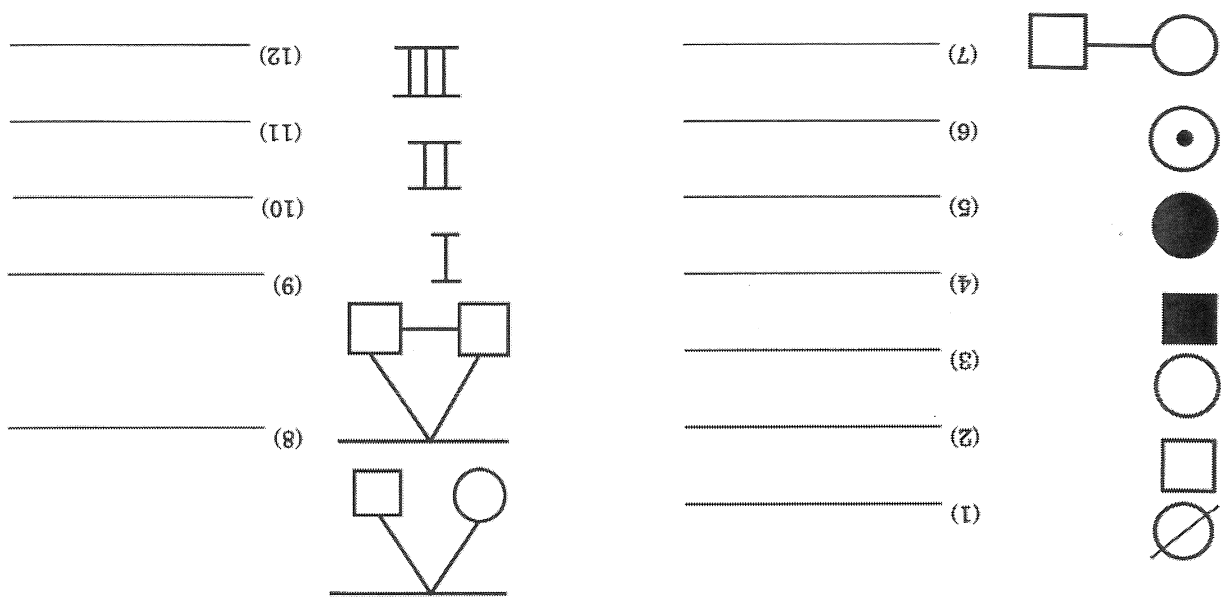
1. Describe the meaning of the term Pedigree.
2. Describe what the various symbols in a pedigree represent.
3. Construct a pedigree showing the family history of a particular trait.
4. Given a pedigree determine if a trait is dominant or recessive.
5. Given a pedigree determine if a trait is autosomal or sex-linked.
6. Describe the mode of inheritance, characteristics and possible treatments, for (a) Huntington's disease, (b) PKU, (c) Duchene muscular dystrophy, (d) Sickle-cell anaemia, (e) Red-green colour blindness.
7. Describe how Mitochondrial DNA is inherited and how it can be used to help study human ancestry.
8. Describe the processes of amniocentesis and chorionic villus sampling.
9. Explain what was involved in the human genome project.

Pedigrees, Genetic Testing and the Human Genome

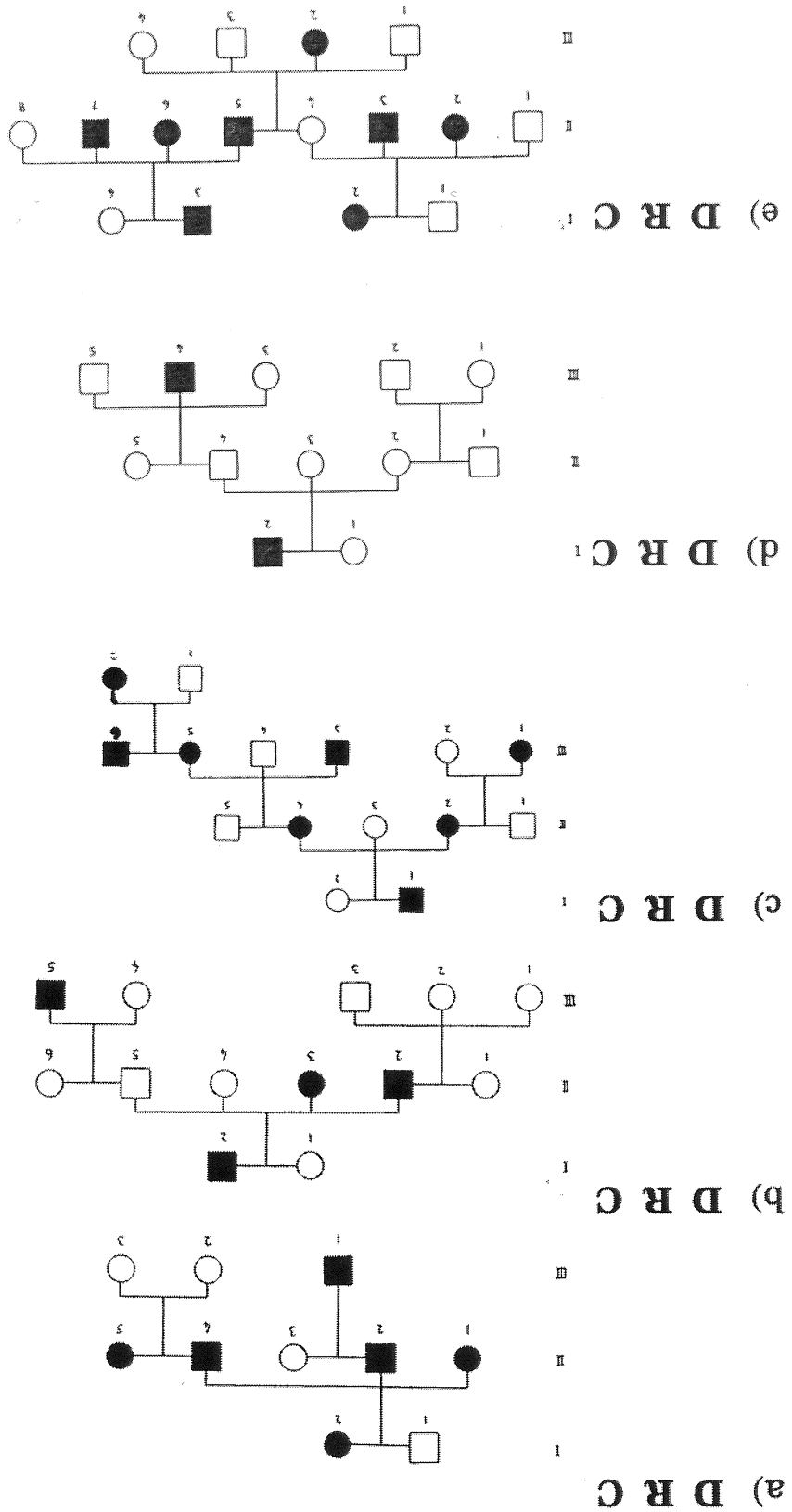
Project Questions

1. What is a Pedigree?

2. Indicate what the following pedigree symbols represent.

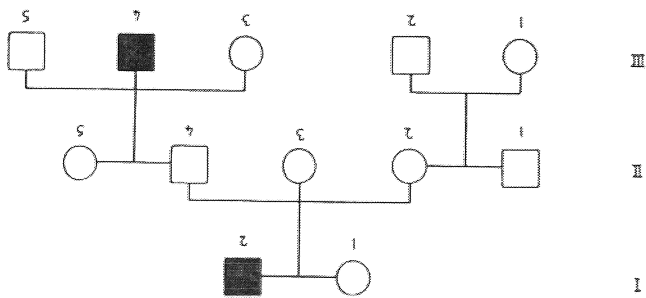
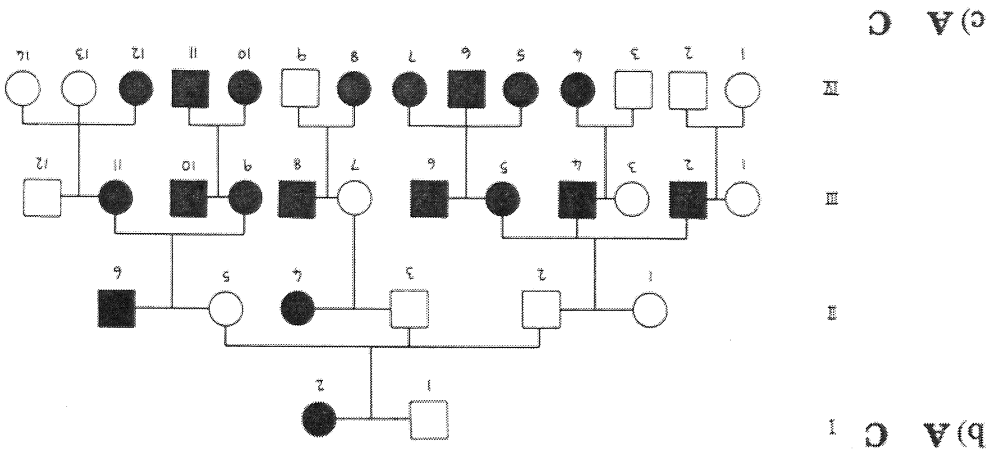


6. Below are a variety of pedigrees for inherited characteristics. In each case, decide whether the trait is dominant (D), recessive (R) or cannot be determined from the information given in the pedigree). Circle the part of the pedigree which allows you to determine dominant or recessive traits.



If a trait is sex-linked then if a mother has the trait then all her sons must have it and if a father does not have the trait then none of his daughters can have it. So when looking at a pedigree to determine if it is not sex linked i.e. autosomal, look for a mother who has the trait and her sons do not and a father without the trait and who has daughters with it.

For the following recessive pedigrees decide if they are autosomal recessive (A) or cannot distinguish between autosomal and sex-linked (C). Circle the appropriate area on the pedigree that indicates if it is autosomal recessive.



13. What was involved in the human genome project?

Pedigrees, Genetic Testing and the Human Genome Project Terminology

1. The more correct term for a "family tree":

2. Affected individuals in a pedigree are indicated by:

3. Deceased individuals in a pedigree are indicated by:

4. A trait determined by genes on non-sex chromosomes:

5. A trait that can be masked or hidden:

6. In a pedigree homozygous recessive individuals will always be:

7. In a pedigree for a sex-linked recessive trait if the mother exhibits the trait then all of her _____ will also exhibit the trait:

8. DNA can be found in the nucleus and in:

9. Genetic testing involving examination of the amniotic fluid:

10. Genetic testing involving examination of the chorion:

11. The international effort to map human DNA:

4. Distinguish between somatic and germ-line mutations.

5. Describe some of the effects of mutations.

6. What is a karyotype and how can karyotypes be used to determine sex an possible genetic disorders.

7. Explain the cause and effect of the following disorders.

Disorder	Cause	Effect
Albinism		
Duchenne Muscular Dystrophy		
Cystic Fibrosis		
Tay-Sachs Disease		
Down Syndrome		
Klinefelters Syndrome		

8. Explain monosomy and give an example.

Mutations, Variation and the Environment

Terminology

1. Term for a spontaneous inheritable change in DNA:
2. Agents that increase the rate of mutations:
3. Term for the process by which better adapted organisms have a greater chance of survival and therefore leave more offspring with their inherited characteristics:
4. Term for the change in a species over time as a result of natural selection:
5. Any characteristic of an organism that gives it a greater chance of survival in its new environment:
6. Term describing all the genes of a population:
7. Term describing organisms that can interbreed, under natural conditions and produce fertile offspring:
8. An organism with a characteristic resulting from a mutation:
9. Changes in a single gene:
10. Mutation where all or part of a chromosome is effected:
11. Mutations that occur in the body cells:
12. Mutations that occur in the gametes:
13. PKU:
14. Change in only one base pair:
15. Absence of pigment from the hair, skin and eyes:
16. Wasting of first leg muscles then arms and chest:
17. Persistent coughing and wheezing:
18. Recessive mutations that cause death:
19. Occurs more frequently in children of older mothers:
20. A photograph of chromosomes in order:
21. Syndrome where males have small testes and breasts (XXY or XYY):
22. Where individual is missing a chromosome of the homologous pair:
23. Differences between individuals:
24. Loss of part of a chromosome:
25. Where section of chromosome occurs twice:

Reproductive Systems Questions

1. Define the following terms;

a) Reproduction –

b) Asexual reproduction –

c) Sexual reproduction –

d) Fertilisation –

e) Internal fertilisation –

2. Reproduction is not necessary for the survival of an individual. Why then do individuals reproduce?

3. How is the genetic information transferred from parent to their offspring?

4. Define the following terms;

a) Gonads –

b) Zygote –

c) Primary sex organs –

d) Secondary sex organs –

5. Describe sexual dimorphism and the reason for it.

7. Describe the general function of each of the parts of the male reproductive system in the table below.

Structure	Function
Ureter	
Bladder	
Pelvis	
Urethra	
Erectile Tissue	
Foreskin	
Testes	
Seminal Vesicle	
Prostate	
Bulbo-urethral Gland	
Vas Deferens	
Epididymis	
Scrotum	
Penis	
Seminiferous Tubules	

9. Describe the general function of each of the parts of the female reproductive system below.

Structure	Function
Fimbriae	
Uterus	
Cervix	
Rectum	
Vagina	
Labia	
Clitoris	
Urethra	
Bladder	
Uterine tube	
Ovary	
Pelvis	
Endometrium	
Anus	
Bladder	

12. Draw diagrams to illustrate the processes of spermatogenesis and oogenesis.

13. How do the processes of spermatogenesis and oogenesis differ?

14. Explain in sequential order the processes that cause an erection following mental or physical stimulation.

24. Describe the timing and changes in structure as a zygote develops into a morula then a blastocyst.

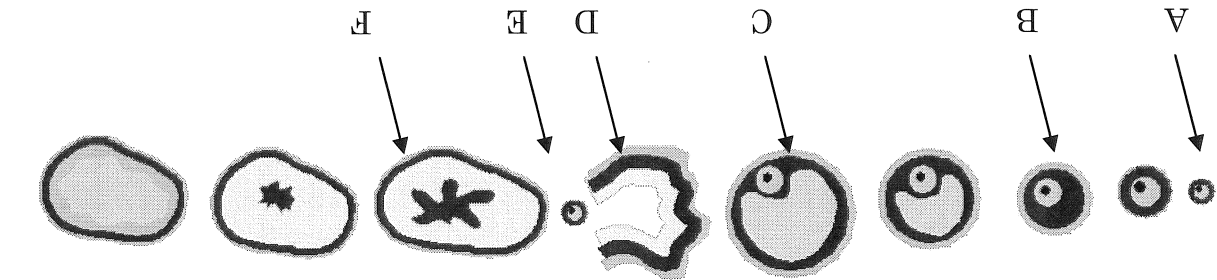
25. Explain how the zygote implants into the endometrium.

26. List five signs and symptoms of pregnancy.

27. Explain how each of the following sets of twins occur;
a) Non-identical – _____

b) Identical – _____

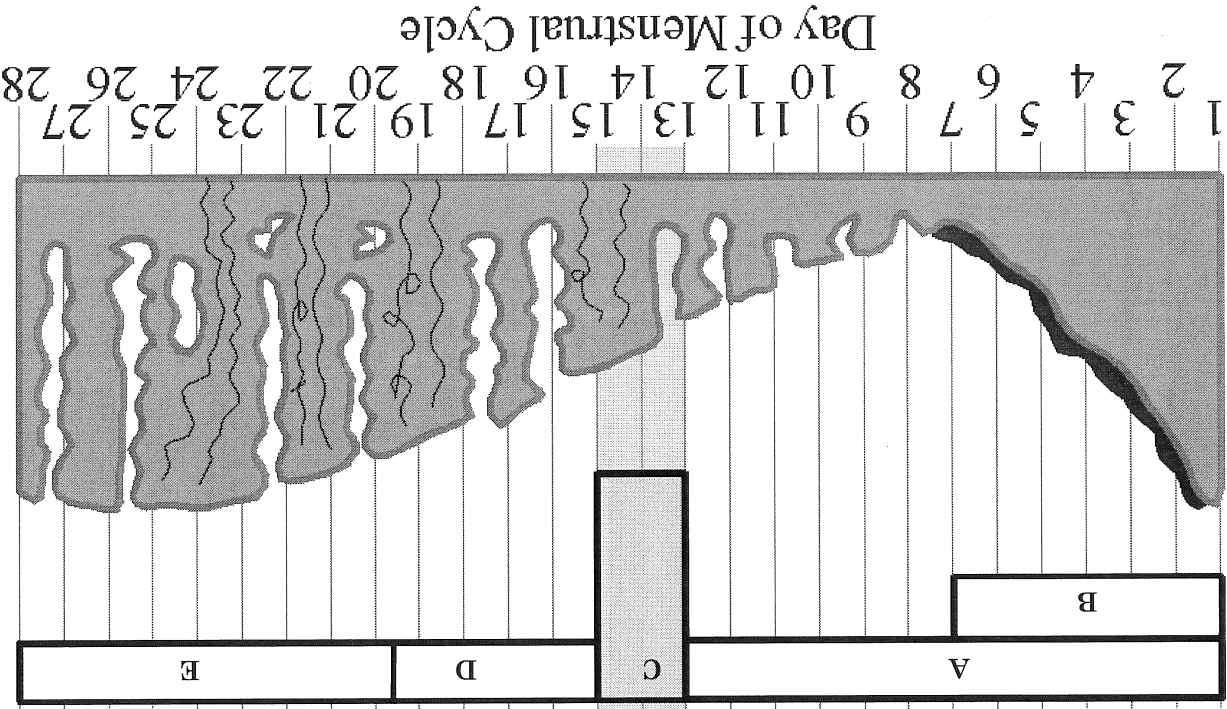
28. Explain the role of endocrine glands and hormones in communication within the body.



32. Label the following diagram of the ovarian cycle with the terms below.
(Mature follicle, Ova, Secondary follicle, Primary follicle, Corpus luteum, Ovulation)

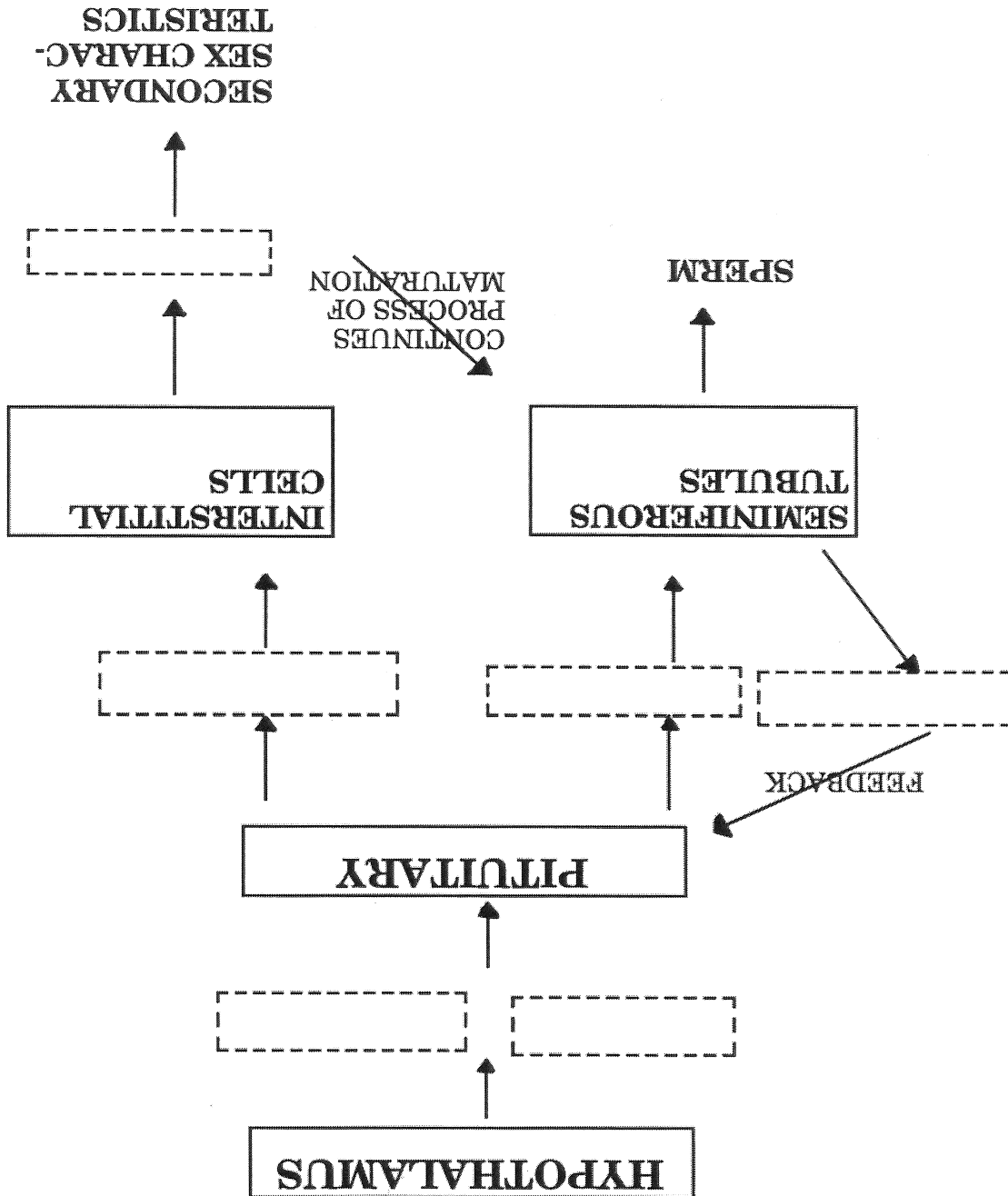
- A - _____
- B - _____
- C - _____
- D - _____
- E - _____
- F - _____

33. Label the following diagram of the menstrual cycle with the terms below.
(Ovulation, Menstruation, Pre-ovulation, Secretion, pre-menstruation.

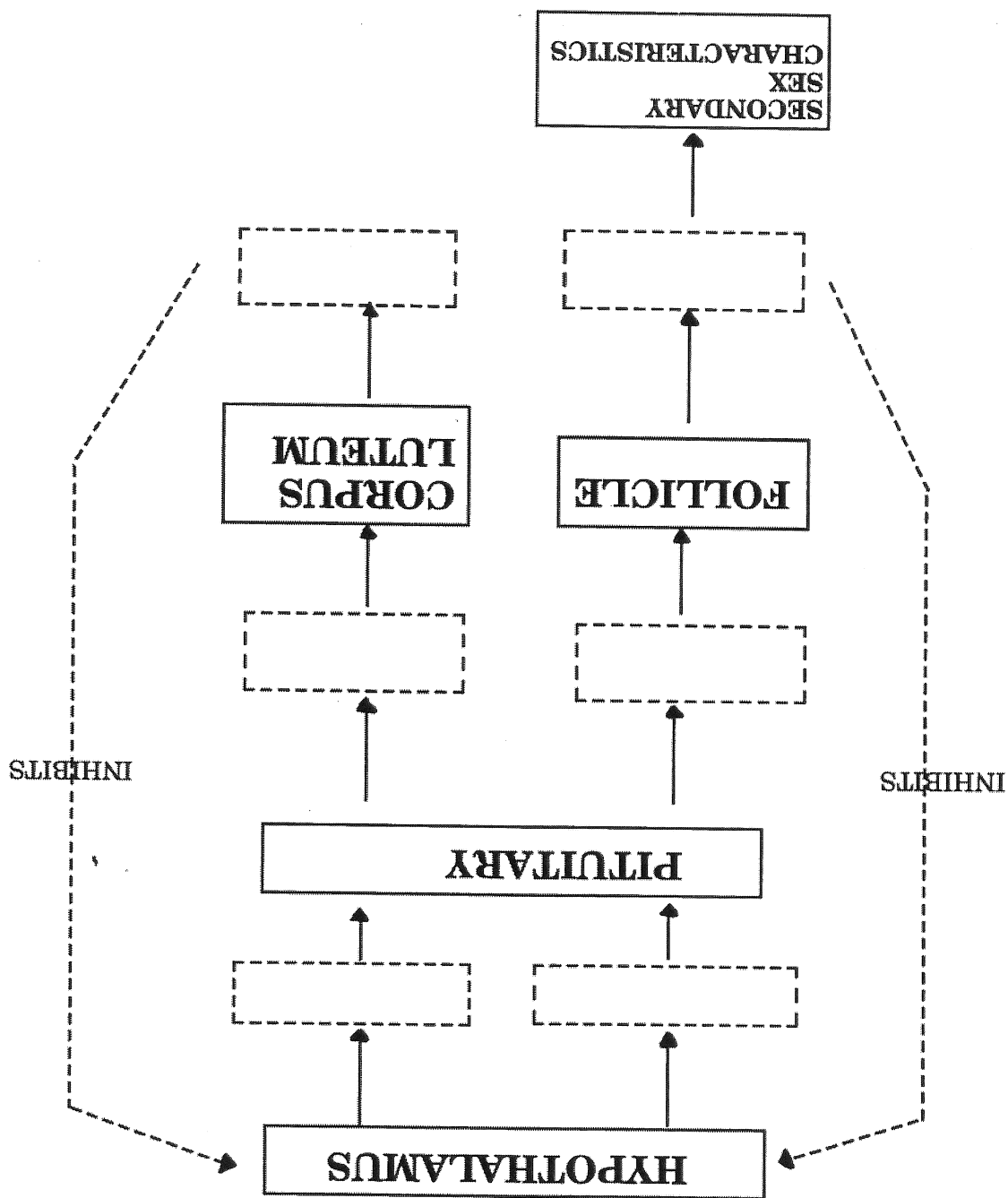


- A - _____
- B - _____
- C - _____
- D - _____
- E - _____

- regulation of spermatogenesis in males.
(FSH-rf, LH-rf, FSH, LH, Inhibin, Testosterone)



37. Use the following list of hormones to complete the diagram below on the hormonal regulation of oogenesis in females.
(FSH-rf, LH-rf, LH, FSH, OESTROGEN, PROGESTERONE)



30. Folds of skin and tissue that have majora and minora parts:

31. The name the describes the number of chromosomes in somatic cells:

32. Cycle of events that takes place in the ovaries:

33. When the female matures sexually:

34. The first menstruation for females:

35. Ceasing of the menstrual cycle:

36. Release of ovum from the ovary:

37. Shedding of the endometrium through the vagina:

38. Hormone that stimulates follicle development:

39. Hormone that causes ovulation:

40. Produces both FSH-rf and LH-rf:

41. Feedback where a rise in one hormone causes a drop in another:

42. Form of cell division that produces gametes:

Differentiation & Development Questions

1. Describe the following structures;

a) Zygote –

b) Morula –

c) Blastocyst –

2. Define the following terms;

a) Cleavage –

b) Implantation –

3. a) Describe what is meant by the term stem cell.

b) Distinguish between the terms proliferation and differentiation.

4. Complete the following table on the potential for development of the various types of stem cells.

Type of stem cell	Totipotent	Pluripotent	Multipotent
Potential to develop (potency)			

10. Describe the major changes in structure during the period of the embryo - week 0-8.

11. Describe the major changes in structure during the period of the foetus - week 9-39.

12. When does each of the following milestones of development occur?

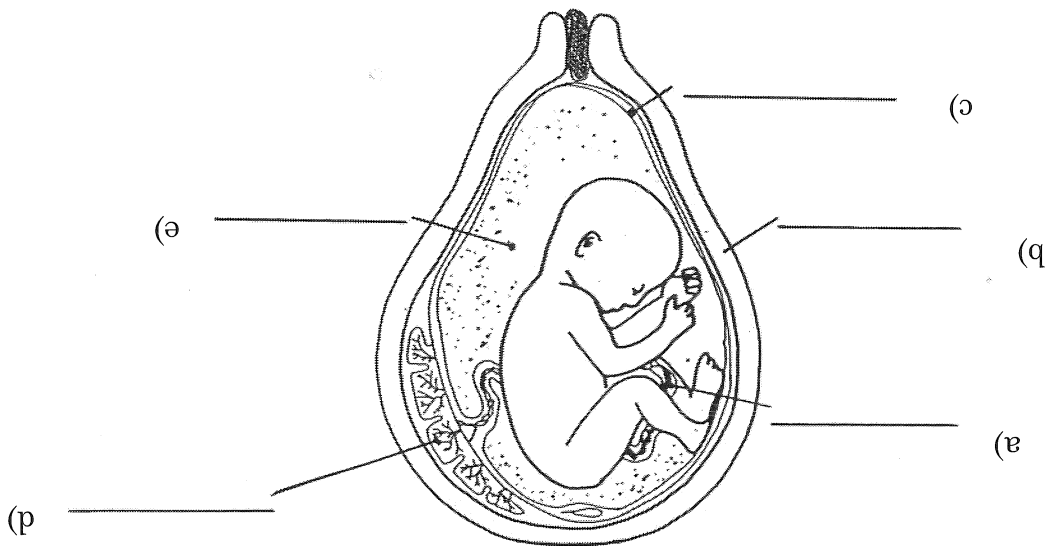
a) Heart beat –

b) Ossification –

c) Limb buds –

13. In what week of development is the developing embryo termed a foetus?

14. Label the following diagram of the embryonic membranes.



18. Describe the general role and structure of the umbilical chord.

19. Outline the physiological changes in the mother 'to be' and the considerations that she must take during pregnancy, under the following headings.

a) Circulatory system –

b) Respiratory system –

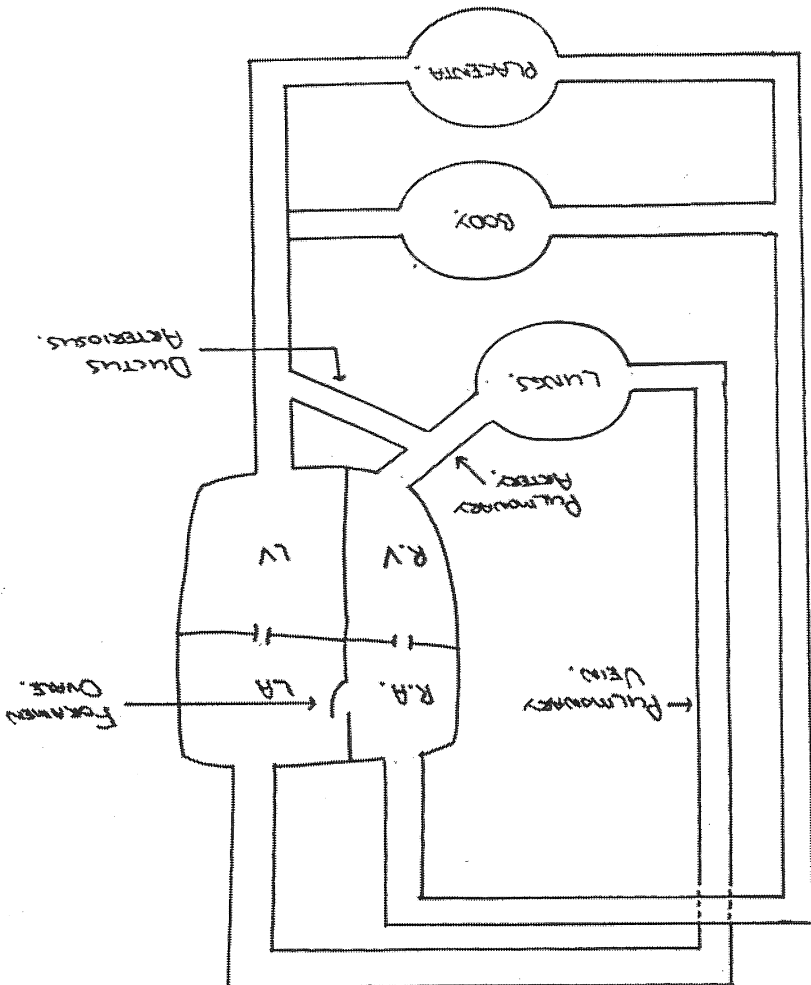
c) Digestive system –

d) Diet –

e) Weight gain –

f) Drugs –

25. On the following diagram of the circulation of a fetus/baby, use a red pen to show the direction of blood flow before the birth and the blue pen to show the direction of blood flow after birth.



26. Explain the following in terms of postnatal care;

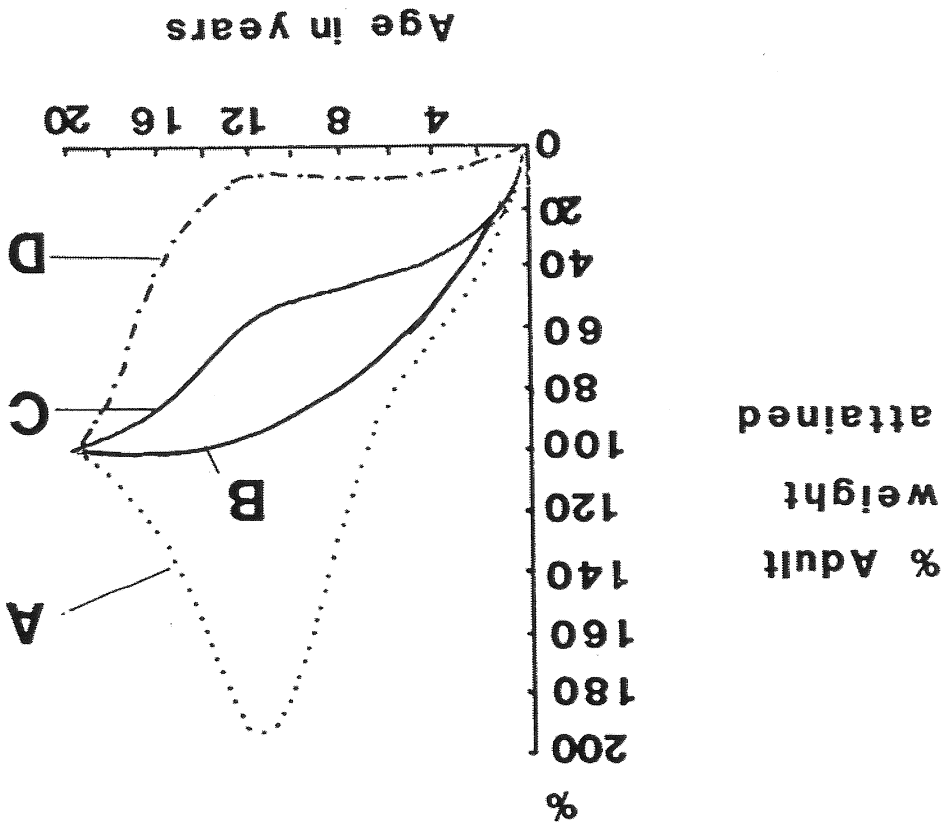
a) Body temperature –

b) Water balance –

c) Infection –

d) Diet/nutrition –

29. Use the words below to label the following graph showing the growth of the various tissues (nervous, general, lymphoid, and reproductive).



30. Explain the reasons for the variable growth in each of the tissues labelled above.

a) Lymphoid –

b) Nervous –

c) Reproductive –

d) General –

35. Define socialisation.

36. List and describe four factors that influence social development.

a)

b)

c)

d)

37. Describe the changes in type and complexity of speech /language as a new born baby develops into a toddler and then a child.

38. The development of speech in children that do not have contact with their parents or other family members is often detrimentally affected. Explain why this is so.

Environmental Factors and Reproductive Technologies Objectives

At the end of this unit students should be able to:

1. Describe the care which must be taken with the unborn child with relation to; smoking and alcohol.
2. Define the term teratogen and describe the effects of a number of teratogens on the unborn child.
3. Explain the following assisted reproductive technologies (ART); in-vitro fertilisation (IVF), gamete intrafallopian transfer (GIFT), intracytoplasmic sperm injection (ICSI), donor ovum or embryo and surrogacy.
4. Define the term contraception.
5. Describe the mechanism of action and give the relative and advantages and disadvantages of a range of contraceptive techniques.
6. Give the causative agent, the effects on the body, control and treatment for the following STI's; Syphilis, Gonorrhoea, Chlamydia, Herpes and AIDS.
7. Describe how ultrasound, foetal monitoring and hormonal intervention are used in the maintenance of pregnancy.

Environmental Factors and Reproductive Technologies Questions

1. Describe why care must be taken with the unborn child with relation to smoking and alcohol.

2. Define the term teratogen and describe the effects of two teratogens on the unborn child.

6. Complete the following table on sexually transmitted diseases infections.

STI	Causative agent	Effects on body	Control	Treatment
Syphilis				
Gonorrhoea				
Chlamydia				
Herpes				
AIDS				

Environmental Factors and Reproductive Technologies Terminology

1. A substance than cause physical defects in an embryo/foetus:
2. The acronym ART stands for:
3. The acronym IVF stands for:
4. The acronym GIFT stands for:
5. The acronym ICSI stands for:
6. The term for a woman who bears a child for an infertile couple:
7. A measure to prevent a pregnancy occurring or continuing:
8. Contraceptive method involving the determination of ovulation and avoidance of intercourse:
9. The condom and diaphragm are examples contraceptive methods that work by forming a:
10. The combined hormonal pill contains the hormones:
11. The acronym IUD stands for:
12. The type of micro-organism responsible for syphilis and gonorrhoea:
13. The sore or ulcer caused by syphilis:
14. The most effective method of preventing STIs:
15. Type of STI that has few symptoms in females and can lead to sterility:
16. Causative agent of AIDS:
17. Virus that can cause blisters on genital organs:

20. (a) $\text{Percentage Change} = (\text{Final value} - \text{Initial value}) / \text{Initial value} \times 100$. (b) Positive 5.9%. (c) Negative 5.8%.

21. Should include if the results "support" or "do not support" the hypothesis. Averages and percentage changes should be quoted where appropriate.

22. (a) "That LACTPLUS will increase milk production in dairy cows by over 20%". (b) LACTPLUS. (c) Milk production. (d) Same age, same breed. (e) Food, water. (f) Group B. (g) 25. (h) "To reduce the influence of biological variation and experimental error. (i) To reduce the effect of bias, to not introduce another variable. (j) Biological variation. (k) Group A averaged 0.9 L / cow / day more than group B. This represents a 21.4% increase. Therefore the hypothesis (and the company's claim) has been supported.

23. (a) "That DISSOLVOPAT will reduce 10% of your body mass in 10 days". (b) DISSOLVOPAT. (c) Body mass. (d) Age, health, breed. (e) Diet, water intake, temperature of room, exercise. (f) Group B. (g) 25. (h) Reduce the influence of biological variation and experimental error. (i) Reduce the influence of bias by introducing another variable. (j) To reduce the influence of the psychosomatic effect. (k) Group A Negative 11.5%, Group B Negative 2%. (l) The results support the hypothesis because Group A the experimental group lost 11.5% on average over the ten day period. (m) Experiment was carried out on rats not humans, experiment was not repeated. Do a trial on human volunteers and repeat the experiment a number of times. (n) Yes to reduce the influence of biological variation and experimental error.

The Practice of Human Biology Terminology Answers

1. Hypothesis. 2. Observations. 3. Technology. 4. Independent. 5. Sample size and replication. 6. Comparison. 7. Bias. 8. Inference. 9. Dependent. 10. Physcosomatic. 11. False. 12. Theory. 13. Test. 14. Biological variation. 15. Extrapolate. 16. Placebo. 17. Continuous. 18. Line. 19. Controlled. 20. Interpolation.

Cell Structure & Mitosis Answers

1. Cell, Tissue, Organ, System, Organism.

2. The structural and function unit of the Human Body.

3. A-Cytoplasm, B-Endoplasmic Reticulum, C-Ribosomes, E-Nucleoplasm, F-Mitochondrion, G-Cell Membrane, J-Nucleolus, K-Nuclear Membrane, L-Golgi Body, O-Centrioles.

4.

Organelles	Function of Organelles
Cell membrane	Differently permeable, controls the movement of materials into and out of the cell.
Mitochondria	Where aerobic respiration takes place. Makes energy available for the cells activities.
Nucleus	Controls the activities of the cell through protein synthesis.
Lysosomes	Contains digestive enzymes that are able to breakdown proteins, lipids and carbohydrates.
Ribosomes	Assembles amino acids to make proteins.
Endoplasmic Reticulum	Increases surface area for chemical reactions. Channels used as a transport system.
Cytoplasm	Liquid medium for cell metabolism. Where organelles are suspended.
Centrioles	Produce spindle fibres. Involved in the process of cell division.
Golgi Apparatus	Concentrates, stores and packages materials for secretion from the cell.
Nucleolus	RNA synthesis for ribosomes.

- 10 a) Enzymes have an optimum temperature usually in humans (37 degrees Celsius). Extremes of temperature denature enzymes. b) pH sensitivity – Enzymes have an optimum pH, extremes of pH will denature an enzyme. c) The higher the enzyme concentration the faster the reaction. d) Changing the substrate concentration has no effect on the rate of the reaction, for a given quantity of enzyme.

11. No answer provided.

12. No answer provided.

13. a) The name ends in –ase i.e. Protease acts on proteins, Lipase on Lipids, etc. b) Lock and Key hypothesis. The shape/structure of the enzyme allows it to combine with the substrate, reducing the activation energy required for the reaction, allowing the reaction to proceed at a faster rate. c) From left to right; Enzyme, Substrate, Enzyme/Substrate Complex, Enzyme and Products. d) Breakdown/Catabolic. e) Yes; however the synthesis reaction would require a greater input of energy. f) Active sites. g) With the active site blocked, the enzyme-substrate complex would not form and the reaction would effectively cease. h) Used in anabolism and catabolism, reusable and specific.

14. a) Changes the shape/structure of the protein of the enzyme (denatures/coagulates the protein). Therefore no longer have the correct active sites. b) The condition of temperature and pH that produce the greatest rates of reaction, about 37 degrees Celsius and pH 7 for most human enzymes.

15. Many enzymes require the presence of certain ions before they will catalyse a reaction. These are called co-factors. Some co-factors are non-protein organic molecules, these are called co-enzymes.

Metabolism Terminology Answers

1. Synthesis. 2. Catabolism. 3. Anabolism. 4. Aerobic. 5. Anaerobic. 6. Mitochondrion. 7. Adenosine Tri Phosphate. 8. Active site. 9. Nutrient. 10. Protein. 11. Minerals. 12. Lipids. 13. Protein. 14. Catalysts. 15. Optimal. 16. Denature. 17. Activation energy. 18. Co-enzyme.

Cell Transport Answers

1. a) A differentially permeable membrane that controls the movement of materials into and out of the cell. Semi-permeable membrane is a membrane that permits the passage of certain substances (usually smaller ones) but restricts the passage of others (usually larger ones). b) Lipoprotein c) i) Denature protein of membrane (ii) Dissolves lipid from membrane (iii) Disrupts structure of membrane; separates lipid/protein. d) Diffusion, active transport, endocytosis and exocytosis.

2. a) Passive processes do not require metabolic energy from the cell to occur. Active processes do require metabolic energy from the cell to occur. b) Endocytosis is the process of taking materials into the cell while exocytosis is the removing of materials from the cell to the external environment. c) i) Passive, ii) Active, (iii) Active, (iv) Active. d) i) Diffusion - The NETT movement of molecules from regions of high concentration to regions of low concentration. ii) Facilitated diffusion - Diffusion of larger molecules through the membrane by carrier proteins. Is a passive process that moves molecules from areas of high concentration to areas of lower concentration. (iii) Osmosis - Osmosis is the diffusion of water. It is the NETT movement of water from an area of high water concentration (dilute solution) to an area of low water concentration (concentrated solution). (iv) Active Transport - The movement of molecules across the cell membrane and AGAINST the concentration gradient, from regions of low concentration to regions of high concentration. This involves the expenditure of metabolic energy by the cell. (v) Pinocytosis – The taking in of liquids by vacuole formation by the cell membrane. (vi) Phagocytosis – The engulfing of larger, solid particles by vacuole formation by the cell membrane. e) i) Random molecular movement, i.e. kinetic energy of molecules. ii) Increase the concentration gradient and increase the temperature. (iii) No, Net diffusion is zero, but the molecules would still be moving equally in both directions across the membrane. f) i) Observation - Red blood cell shrinks, Explanation – Net movement of water out of the red blood cell into the honey. ii) Observation - Potato cube swells, Explanation – Water diffuses into potato cube. (iii) Observation - Red blood cell bursts, Explanation – Net movement of water into the red blood cell from the distilled water. (iv) Observation – Potato cube shrinks. Explanation – Water diffuses out of potato cube. (v) Observation – Level in side A falls, B rises. Explanation – Net diffusion of water from side A to side B.

7. No answer supplied.
8. Atria only have to pump blood short distance to ventricles, ventricles have to pump blood larger distances to lungs and body.
9. Left ventricle has to pump blood a large distance throughout the whole body, right ventricle has to pump blood a shorter distance to lungs.
10. Pericardium- Encloses and holds heart in place, prevents heart from over-stretching allows for friction to be reduced when beating.
11. No answer supplied.
12. a) Systemic- Transports nutrients and oxygen to body cells and wastes and carbon dioxide away.
b) Pulmonary- transports deoxygenated blood to lungs where oxygenation takes place, and returns oxygenated blood to the lungs.
c) Portal- Transports blood containing nutrients absorbed from the gut to the liver for storage.
13. Left.
14. Right.
15. Prevents mixing of oxygenated and deoxygenated blood thus allowing a more efficient oxygen transfer in the circulatory system.
- 16.
- | | | | | |
|---------------------|-----------------------------------|------------------------------|-----------------------|----------------------|
| Blood vessel | Relative thickness of wall | Valves present/absent | Muscle present | Size of lumen |
| Arteries | Thick | Absent | Present | Small (changeable) |
| Veins | Thin | Present | Absent | Large |
| Capillaries | 1 cell thick | Absent | Absent | Very small |
17. a) Arteries- Carry blood away from the heart (all arteries carry oxygenated blood except the pulmonary).
b) Veins- Carry blood to the heart (All veins carry deoxygenated blood except for pulmonary).
c) Capillaries- Very close to all cells, where transfer of nutrients, gases and wastes takes place between cells and blood.
18. a) Capillary. b) Artery. c) Vein.
19. Pulmonary Arteries, carry deoxygenated blood to lungs.
20. The muscle can contract or dilate the lumen thus increasing or decreasing the amount of blood flowing through the artery.
21. When the ventricles contract the walls of the arteries expand (stretch) due to the blood pressure. When the heart relaxes the elastic recoil of the artery walls keeps the blood moving by maintaining the blood pressure. By maintaining the blood pressure a constant flow of blood is maintained to the capillaries.
22. The blood in the veins is under low pressure. The valves prevent back flow of blood, the large lumen reduces frictional resistance of the vein walls on the blood. The veins close proximity to skeletal muscles means that the contraction of these muscles helps 'push' blood along the veins.
23. Allows easy and fast transport of substances by diffusion, between the cell intercellular fluid and plasma.
24. a) Arterioles. b) Venules
25. a) Plasma. b) Erythrocytes (red blood cells). c) Leucocytes (White blood cells). d) Thrombocytes (Platelets)

40. a) The term given to the condition where there is an insufficient blood supply to the body's vital tissues. b) External or internal loss of blood (bleeding).

41. 1. Apply direct pressure to an area to stop/reduce blood loss. 2. Lay patient down in prone position. 3. Elevate wounded body part. 4. Immobilise body part that is bleeding. 5. Treat for shock. 6. Seek medical aid.

42. Redness, swelling, pain & heat.

43. Histamine increases the flow of blood to the infected area. This causes the redness and heat. Histamine also increases the permeability of the capillaries. This results in more plasma entering the tissues which results in swelling of the infected area.

Circulatory System terminology answers

1. Red. 2. Atria. 3. Ventricles. 4. Pulmonary arteries. 5. Oxygenated. 6. Atrioventricular valves. 7. Semilunar valves. 8. Pulmonary. 9. Left. 10. Left. 11. Mesenteric. 12. Carotid. 13. Coronary. 14. Artery. 15. Veins. 16. Smooth. 17. Venules. 18. Plasma. 19. Erythrocytes. 20. Leucocytes. 21. Thrombocytes. 22. Biconcave discs. 23. Red bone marrow. 24. Spleen. 25. Diastole. 26. Sphygmomanometer. 27. Systolic. 28. 120/80 mmHg. 29. Lymph. 30. Shock. 31. Haemorrhage. 32. Histamine. 33. Pus

Respiratory System Answers

1. Individual cells of the organism are too far removed from the external environment for gas exchange to occur (efficiently enough) by diffusion.
2. A) Tracheal Cartilage. B) Trachea. C) Intercostal Muscle. D) Rib. E) Left bronchus. F) Pleural cavity. G) Alveoli. H) Bronchiole. I) Inner pleural membrane. J) Outer pleural membrane. K) Diaphragm. L) Nasal cavity. M) Epiglottis. N) Larynx. O) Pharynx.
3. a) No answer supplied b) Nasal cavity → Larynx → Trachea → Bronchus → Bronchial tree → Bronchiole → Alveoli
4. Tracheal Cartilage: Prevents collapsing of the trachea. Trachea: Conducts air from pharynx/larynx to bronchi. Intercostal Muscle: Raise ribs to increase volume of thoracic cavity. Rib: Protection of thoracic organs (Heart/Lungs). Left bronchus: Conducts air from trachea to bronchial tree. Pleural cavity: Contains pleural fluid that allows the pleural membranes to slide over each other reducing friction. Alveoli: Surface for gas exchange between blood in capillaries and alveolar air. Bronchiole: Conduct air from bronchial tree to alveoli they control the amount of air entering the alveoli by a layer of smooth muscle. Pleural membranes: Pleural membranes together with the pleural fluid hold the outside of the lungs against the inside of the chest wall. Diaphragm: Flattens during contraction to increase the volume of thoracic cavity. Nasal cavity: Warms, moistens (humidifies) and filters the air. Epiglottis: Covers trachea during swallowing to prevent food/liquid passing into the trachea. Pharynx: Air from nasal cavity passes through. Larynx: Production of sound that leads to speech. Lung: Pair of organs in the chest cavity comprised of tubes and air sacs used for gas exchange.

5. a) or ventilation; is the process of moving air into and out of the lungs. b) or inhalation; is the process of drawing air into the lungs it is an active process involving the contraction of muscles. c) or exhalation; is the process of forcing air out of the lungs it is a passive process and is due to the relaxation of the relevant muscles. d) The process of releasing useful energy from organic compounds usually with the use of oxygen. It occurs in cells.

6. a) Contract b) rise, out c) Contracts, down d) forward e) increasing, decreasing, into
7. a) Relax. b) Fall down. c) Relaxes, up. d) Back. e) Decreasing, increasing, out.

Lung cancer	Bronchitis	Carbon monoxide poisoning	Pneumonia
Uncontrolled division of cells within the respiratory system resulting in formation of tumour.	An increase in mucus secretion in Bronchus/ infection. Exposure to irritants.	Reduction of oxygen carrying capacity of blood due to carbon monoxide combining with haemoglobin more readily.	Infection/ Inflammation of the air sacs in the lungs.
Exposure to pollutants, irritants and carcinogens.	Microbiological infection. Exposure to irritants.	Carbon monoxide produced by burning of fuels.	Caused by virus or bacteria.
Changes in respiratory functions caused by tumour. Blood in sputum.	Increase in mucus secretion results in Bronchus/Bronchioles accumulating mucus affecting ventilation.	Drowsiness Decrease in consciousness Death due to insufficient oxygen.	Shortness of breath, fever, chest pain.
1.Surgery. 2. Radiotherapy. 3. Chemotherapy.	1.Antibodies. 2. Drugs. 3. Physiotherapy. 4.Coughing to clear built up.	1. Removal from source. 2. Administration of oxygen.	1. Antibiotics. 2. Fluid replacement.

Respiratory System Terminology Answers

1. Respiratory. 2. Ventilation. 3. Nasal cavity. 4. Cilia. 5. Olfactory. 6. Mucus. 7. Pharynx. 8. Larynx. 9. Vocal cords. 10. Loudness. 11. Pitch. 12. Epiglottitis. 13. Trachea. 14. Bronchi. 15. Bronchioles. 16. Inspiration. 17. Expiration. 18. Pleural. 19. Intercostal. 20. Diaphragm. 21. Alveoli. 22. Ribs. 23. Carbon Dioxide. 24. Oxyhaemoglobin. 25. Bicarbonate ion. 26. Carboxyhaemoglobin. 27. Bright red. 28. Lung. 29. Emphysema. 30. Pneumonia. 31. Asthma. 32. Diffusion. 33. Concentration gradient. 34. Oxygen. 35. Deoxygenated. 36. True. 37. Decrease. 38. Down. 39. Danger. 40. Oxygen.

Digestive System answers

1. a) The act of taking food into the alimentary canal through the mouth.
b) The process by which insoluble food (large particles) is broken down into soluble components (small molecules).
c) The expulsion from the alimentary canal of the undigested remains of food.
d) The break down of food by the action of the mouth (teeth, tongue lips) and the churning action of the alimentary canal into smaller particles. Decrease in particle size and increase surface area.
e) The breakdown of food by digestive juices and enzymes into soluble components.
2. Mechanical digestion includes the physical break down of food into smaller particles. Most mechanical digestion takes place in the mouth the teeth act in combination with the lips and tongue. The various teeth cut, bite, crush and grind food into smaller particles. Mechanical digestion also occurs in the stomach by the churning of the food (chyme) due to muscular contraction.
3. a) Incisors. b) Canines, c) Premolars. D) Molars.

4.

Tooth type	Total number	Description	Function
Incisors	8 (4 in each jaw)	• Chisel shaped crown • 1 root	Cutting and biting
Canines	4 (2 in each jaw)	• Conical shaped crown • 1 root	Ripping and tearing
Premolars	8 (4 in each jaw)	• Square shaped crown with 4+ cusps • 2+ roots	Chewing, crushing and grinding
Molars	12 (6 in each jaw)	• Square shaped crown with 4+ cusps • 2+ roots	Chewing, crushing and grinding

14. a) Mouth: Mechanical digestion (teeth) and chemical digestion (salivary glands).
 b) Salivary Glands: Production of saliva, lubrication and amylase.
 c) Pharynx: Joins mouth and oesophagus.
 d) Oesophagus: Connects pharynx and stomach.
 e) Epiglottis: Covers trachea during swallowing.
 f) Teeth: Mechanical digestion cut, rip, bite, tear, crush, grind and chew food into smaller pieces.
 g) Tongue: Moves food around mouth during mastication.
 h) Stomach: Site for mechanical digestion, chemical digestion of proteins, absorption of some substances.
 i) Duodenum: Site of chemical digestion, secretion of bile and pancreatic juice.
 j) Jejunum: Site of chemical digestion and absorption of soluble food molecules.
 k) Ileum: Site of absorption of soluble food molecules.
 l) Large intestine (Colon): Absorption of water.
 m) Appendix: No specific function, vestigial organ.
 n) Rectum: Stores faeces, muscular movements during defecation, reabsorption of water.
 o) Anus: Final opening of the alimentary canal, site of defecation.
 p) Gall Bladder: Stores bile and secretes the bile into the duodenum.
 q) Pancreas: Stores digestive enzymes and hormones insulin and glucagon.
 r) Liver: Stores glucose as glycogen, produces bile.
 s) Caecum: Joins small intestine to large intestine- no specific function.
15. a) Mucosa, b) Submucosa, c) Peritoneum, d) Muscle, e) Villi, f) Epithelial cell, g) Capillary network, h) Central lacteal, i) Blood vessel.
16. a) Long length: increased time of contact between chyme and intestinal wall.
 b) Large surface area due to villi, microvilli and circular folds and long lengths.
 c) Thin, moist walls to facilitate diffusion.
 d) Capillary network: Efficient capillary network continually removes absorbed nutrients, Lacteal for fats/fatty acids/ Glycerol.
17. a) Great length (6-7 metres).
 b) Circular folds of mucosa.
 c) Villi (small finger like projections of the mucosa).
 d) Microvilli (microscopic hair like projections on villi).
18. a) These are absorbed by diffusion and active transport from the lumen into the villi and pass into the capillary. Active transport is used as it can take place against a concentration gradient and is faster and more complete than diffusion.
 b) A small amount of water is absorbed by osmosis into the villi and passes into the capillaries of the villus. This is a passive process.
 c) Fatty acids along with glycerol are absorbed by diffusion into the villus. They then recombine to form a triglyceride (fat). This is then coated in a layer of protein to form a chylomicron. This then enters the lacteal of the villus.
19. a) The release of a substance (juice/enzyme) from a cell or gland into the alimentary canal.
 b) The movement of small soluble food molecules from the lumen into the blood/lymph.
 c) The rhythmic muscular contraction within the alimentary canal that propels food/chyme along.
 d) The emptying of the faeces contained in the rectum through the anal sphincter.
20. a) A diet high in salt is a contribution factor towards hypertension (high blood pressure). Hypertension itself is a contributing factor towards a number of cardiovascular diseases.
 b) A diet low in soluble or insoluble fibre can contribute to a number of diseases/disorders ranging from diverticulitis to haemorrhoids and bowel cancer.
 c) A diet high in refined sugars can lead to conditions varying from tooth decay and obesity to the aggravation of diabetes.
 d) A diet high in saturated fat can lead to a number of conditions including cardiovascular disease to obesity and bowel cancer.
 e) A diet that results in an excess of kilojoules to that needed to maintain body function, can result in the person becoming overweight or obese.

8. a) Removal of wastes like urea from blood plasma. b) Control of water levels in blood plasma. c) Control of electrolyte levels (Na^+ , Ca^{2+} , Cl^- ect). d) Helps control pH levels.
9. a) Right kidney. b) Renal vein. c) Ureter. d) Aorta. e) Renal artery. f) Left kidney. g) Inferior vena cava. h) Bladder. i) Urethra.
10. a) Supplies blood to kidney. b) Takes blood away from kidney c) Carries urine to bladder. d) Takes urine from bladder to the outside of the body. e) Stores urine.
11. a) Capsule. b) Pelvis c) Calyx. d) Cortex e) Renal artery f) Renal vein g) Ureter. h) Medulla (pyramids).
12. 1) Capsule. 2) Cortex. 3) Medulla. 4) Calyx. 5) Pelvis. 6) Ureter. 7) Renal vein. 8) Renal artery.
13. a) Afferent arteriole. b) Glomerulus. c) Filtration capsule (Bowmans). d) Proximal convoluted tubule. e) Capillary. f) Loop of Henley. g) Efferent arteriole. h) Venule (leads to renal vein). i) Distal convoluted tubule. j) Collecting duct.
14. a) Glomerulus - Works in conjunction with the filtration (Bowmans) capsule to filter the blood. Large substances like red and white blood cells and proteins (which are too large to pass through the membranes of the glomerulus and capsule) remain in the bloodstream and small molecules like water, glucose, salts, amino acids, urea, hormones, etc pass through the membrane into the capsule. b) Glomerular Capsule- as above. c) Proximal Convoluted Tubule - Reabsorption (selective by active transport) of glucose, amino acids, sodium, potassium plus water by osmosis (passive) from the tubule back into the bloodstream. d) Loop of Henle - Reabsorption (selective by active transport) of sodium into tissue fluid to help maintain conditions suitable for the removal of water by osmosis as it moves down the collecting duct. e) Distal Convoluted Tubule - Reabsorption (selective by active transport) of sodium, secretion of hydrogen ions, potassium ions and creatinine into tubule. *The movement of hydrogen ions helps control pH of the plasma and thus other body fluids. f) Collecting Duct - Reabsorption of water into bloodstream by osmosis under the influence of A.D.H., the high sodium concentration outside of duct helps the water move by osmosis. g) Afferent Arteriole - Supplies the nephron with blood and forms a ball of capillaries called the glomerulus. h) Efferent Arteriole - Takes blood away from glomerulus, is smaller in diameter than the afferent arteriole raising blood pressure in the glomerulus, breaks up into the capillary network that surrounds the tubules. i) Capillary Network - Surrounds tubules and duct so that substances that are required by the body in the tubules can be reabsorbed back into the blood stream.
15. Washing hand before preparing food, eating and providing first aid. Washing hand after sneezing, coughing, using the toilet. Avoiding sneezing or coughing over other people, surfaces or food. Avoiding contact with body fluids such as blood. Avoiding contact with objects such as used needles. Not sharing items with others that may contain traces of body fluids or tissue.
16. The skin is a flexible but tough barrier. The layer of dead epidermal cells prevents microbes from entering. The constant shedding of cells dislodges microbes that do attempt to enter. Sweat and oils contain lysozyme that provides a chemical barrier to microbes.
17. A topical preparation is a liquid or cream such as an ointment or gel that is applied to the skin. Topical preparations can be used to protect the skin eg sun cream, prevent infections eg antiseptic cream, or improve the appearance of the skin eg cosmetics.
18. a) 1) Trachea/bronchi – narrow and curved passages. 2) Nasal cavity – Hairs and mucus. 3) Skin – Imperious barrier. 4) Urethra – Flow of urine. 5) Ear – sticky wax. 6) Tears – Flushing effect. 7) Mouth – Saliva washes away microbes. 8) Mucous membranes – trap microbes. b) 1) Ear – wax inhibits growth of microbes. 2) Stomach – HCl kills microbes. 3) Vagina – Acidic environment inhibits microbes. 4) Eyes – lysozyme in tears inhibits microbes. 5) Saliva – lysozyme inhibits microbes.

a particular characteristic (like hair colour) however (ii) functionally they may be different as they may code for a different colour of hair.

7. Meiosis occurs in the testes and ovaries and is involved in the formation of gametes. Four cells are produced which are haploid and contain 23 chromosomes. Mitosis occurs anywhere in the body where growth and repair takes place. It produces two daughter cells which are diploid and contain 46 chromosomes.

8. a) During metaphase 1 the 'legs' of chromosomes become tangled or cross over each other. These points are termed chiasma, sometimes the DNA at these chiasma breaks and chromatids from different chromosomes rejoin. This results in unusual combinations of genes being formed. b) During metaphase 1 when homologous pairs of chromosomes line up there are multiple ways or combinations that can occur. These different combinations result in different combinations of alleles in the gamete. c) Sometimes in anaphase 1 and anaphase 2 the spindle fibres break. This results in more than one chromosome or chromatid ending up in the final cell produced. This results in some cells having excess genes and other lacking particular gene combinations. d) Due to crossing over, random assortment and non-disjunction there are many different 'types' of gametes. During fertilisation there is no way of determining which particular sperm fertilises the ovum. The many different types of sperm can result in many different types of zygotes being produced.

9. a) $23 + 23 \rightarrow 46$ (b) $Y + X \rightarrow XY$, $X + X \rightarrow XX$.

10. a) alleles. b) When a characteristic is controlled by alleles of only one gene pair. c) Phenotypes are in distinct groups (little variation). (d) ABO blood groups, tongue rolling, PTC, albinism, ear lobes, plus others.

11. Dominant alleles 'over-ride/mask' the effect of recessive alleles. In a heterozygous genotype the dominant allele will express itself and the recessive allele will be masked.

12. Heterozygous – Genotype has a pair of different alleles for a characteristic. Homozygous – Genotype has a pair of similar alleles for a characteristic.

13. Capital, Lower case.

14. a) i) BB, ii)bb, iii) Bb. (b) If the person is heterozygous then the genotype will be (Bb) and as the brown gene is dominant the brown eye colour will be expressed. c) i) Bb- 50% B, 50% b. ii) BB- 100% B. iii) bb- 100% b.

15. a) F1 Genotypes 100% Bb, F1 Phenotype 100% brown. b) F1 Genotype 50% BB, 50% Bb, F1 Phenotype 100% brown. c) F1 Genotypes 50% Bb, 50% bb F1 Phenotypes 50% brown, 50% blue. d) F1 Genotypes 25% BB, 50% Bb, 25% bb F1 Phenotypes 75% brown, 25% blue e) This is only an expected outcome, variations may occur due to chance random fertilisation.

16. F = free, f = attached. man = Ff, woman = ff. Genotypes- 50% Ff, 50% ff. phenotypes- 50% free, 50% attached.

17. A = astigmatism, a = normal. man = aa, woman = Aa. Genotypes- 50% Aa, 50% aa. Phenotypes- 50% astigmatism, 50% normal.

18. a) Rr b) rr c) rr d) rr e) Rr.

19. a) A woman homozygous for straight hair. Most chance of producing straight haired children if man is heterozygous, this is an example of a test cross.

20. The alleles are equally dominant. Both alleles are expressed in heterozygous genotypes.

21. a) 1. Homozygous. 2. Haemoglobin type A. 3. Heterozygous. 4. Haemoglobin type A and S. 5. Homozygous. 6. Haemoglobin type S. (b) (i) F1 Genotypes 50% Hb^AHb^S and 50% Hb^AHb^A, F1 Phenotypes 50% type A, 50% type A and S, 25% (ii) F1 Genotype 25% Hb^AHb^A, 50% Hb^AHb^S, 25% Hb^SHb^S. F1 Phenotypes 25% type A, 50% type A and S, 25% type S. (c) Advantages; (i) makes individuals more resistant to malaria. Disadvantages: (i) blood can't carry as much oxygen. (ii) Hb^AHb^S x Hb^AHb^S produces some Hb^SHb^S offspring that die at an early age.

8. a) rr, Red. b) RR or Rr, Normal. c) rr, Red. d) Rr, normal.

9. Mitochondrial DNA is formed into 5-10 small circular molecules that together carry 37 genes.

10. A persons mitochondrial DNA is inherited solely from their mother. The nucleus of the fertilised egg or zygote contains DNA from the father via the sperm and the mother. The mitochondria within the ovum are only from the mother. This allows geneticists to trace peoples ancestry.

11. Amniocentesis involves the insertion of a needle into the amniotic cavity to remove a sample of amniotic fluid. Cells from the foetus floating in the fluid can be examined for abnormalities in chromosome number and structure. The cells can also be screened for biochemical problems. Amniocentesis can be used to identify disorders such as PKU, muscular dystrophy, Down syndrome and cystic fibrosis.

12. Chorionic villus sampling involves the removal of a small part of the chorion foetal membrane and an examination of the cells. Its advantage over amniocentesis is that it can be carried out from 9 weeks and it is easier and faster to complete.

13. Scientists from all over the world cooperated to map all of the genes on the 46 human chromosomes. It took 13 years and mapped the sequence of the bases that make up the human genome

Pedigrees, Genetic Testing and the Human Genome Project Terminology Answers

1. Pedigree. 2. Shading. 3. A line through the symbol. 4. Autosomal. 5. Recessive. 6. Shaded. 7. Sons. 8. Mitochondria. 9. Amniocentesis. 10. Chorionic villus sampling. 11. Human Genome Project.

Mutations, Variation and the Environment Answers

1. A mutation is a sudden permanent change in the bases that make up DNA. An organism that has observable characteristics caused by the mutation is termed a mutant.

2. In a gene mutation only the bases of a particular gene are affected. In a chromosome mutation all or part of a chromosome is involved meaning that many genes are affected.

3. Mutagenic agents or mutagens increase the normally slow rate of mutations. Common mutagens include chemicals eg formaldehyde and ionising radiation eg x-rays.

4. Somatic mutations are mutations that occur in body cells. Only the cell where the mutation occurred and cells that arise from it are affected. Germ-line mutations occur in the gonads and as a result can be passed onto offspring.

5. Mutations can be (i) silent eg there is no observable effect in the phenotype; (ii) beneficial eg a mutant is produced that has an altered structural, functional or behavioural characteristic that is advantageous to its survival; (iii) negative eg the mutant has characteristics that are disadvantageous; (iv) lethal, the changes are so negative that the organism does not survive.

6. A karyotype is a photo/drawing of the full complement of a cells chromosomes arranged in homologous pairs. Karyotypes are used to determine sex by observing the appearance of the sex chromosomes eg XX; female, XY; male. Karyotypes can also indicate the presence of disorders due to extra or missing chromosomes.

2. All organisms have a limited lifespan therefore reproduction is important for the survival of the species.
3. The passing of genes on chromosomes in gametes from parents to offspring.
4. a) Primary sex organ (testis or ovary). b) First cell produced by the combination of sperm and ovum. c) Sex organs that produce gametes. d) Organs required for reproduction under natural conditions but do not produce gametes.
5. a) Physical differences in structure between sexes. So sexes can be easily distinguished.
6. A) Ureter. B) Bladder. C) Pelvis. D) Urethra. E) Erectile tissue. F) Foreskin. G) Testis. H) Seminal Vesicles. I) Prostate. J) Bulbo-urethral Gland. K) Vas deferens. L) Epididymis. M) Scrotum.
7. a) Ureter – transports urine from the kidney to the bladder. b) Bladder – stores urine c) Pelvis – protects and supports some reproductive structures. d) Urethra – carries both semen and urine to the exterior through the penis. e) Erectile tissue – contains sponge like spaces which fill with blood during arousal making the penis erect. f) Foreskin – protection of the head of penis (glans) g) Testis – production of both sperm and testosterone h) Seminal vesicles – secrete a fluid which provides nutrition for and helps to transport sperm. i) Prostate – secretes an alkaline fluid that gives the semen its milky colour neutralises acids from urine and vaginal secretions. j) Bulbourethral Gland – secretes a fluid that acts to lubricate the urethra. k) Vas deferens – conveys sperm from the epididymis to the urethra. l) Epididymis – Storage of the sperm while they mature. m) Scrotum – holds and protects testes also raises and lowers the testes to keep them at the correct temperature. n) Penis – when erect is placed inside the vagina and helps to transport semen from the male to the female. o) Seminal vesicles – Make up testes and are the site of sperm production.
8. A) Fimbriae. B) Endometrium. C) Cervix. D) Rectum. E) Vagina. F) Labia. G) Clitoris. H) Urethra. I) Bladder. J) Uterine Tube. (Fallopian tube) K) Ovary. L) Pelvis. M) Uterus. N) Anus.
9. a) Fimbriae – contract to help to guide ovum into the uterine tube. b) Uterus – place for growth and development of the embryo. Contracts during labour to push out the baby. c) Cervix – joins uterus and vagina, sperm passes through here to get to the fallopian tube. Holds developing foetus in the uterus. d) Rectum – stores faeces (reabsorb water). e) Vagina – accepts penis during intercourse and birth passage. f) Labia – protect vagina and produce fluids which reduce friction during intercourse. g) Clitoris – helps produce sexual arousal when stimulated. h) Urethra – transports urine from bladder to exterior. i) Bladder – stores urine. j) Uterine tube – transports ova from ovary to uterus. k) Ovary – produces and releases ovum, site of production of oestrogen and progesterone l) Pelvis – protects some internal reproductive structures. m) Endometrium – inner lining (soft mucus) of the uterus, where blastocyst implants. n) Anus – sphincter muscle that controls the release of faeces from the anus. o) Bladder – stores urine.
10. a) Form of cell division that is responsible for the production of gametes – four haploid cells produced. b) Production of gametes which contain half the normal chromosome number. c) Production of sperm in seminiferous tubules. d) Production of ova in ovary. e) Cell containing half the normal number of chromosomes (n). f) Cell where chromosomes exist in pairs, the normal number of chromosomes in a somatic cell (2n).
11. No answer supplied.
12. No answer supplied.
13. Spermatogenesis. (i) Takes place in testis. (ii) Takes place continually after puberty. (iii). Four sperm produced per oogonia. (iv) Takes place every 25 days after puberty. (iii) One ovum produced per oogonia
14. Sexual stimulation (mental, physical) → increased blood supply to erectile tissue → increased blood pressure in penis → penis becomes rigid.
15. Semen consists of sperm produced in the seminiferous tubule and seminal fluids produced by the Seminal vesicles, Prostate gland and Bulbo-urethral gland.

30. The menstrual cycle occurs in the uterus and is the breakdown and build-up of the endometrium over 28 days. The ovarian cycle occurs in the ovary and is the development of follicles, ova and a corpus luteum over approximately 28 days.

31. Menarche happens at some stage during puberty and is the start of menstruation. Menopause is the time in women when they cease to menstruate.

32. a) Primary follicle. b) Developing follicle. c) Mature follicle. d) Ovulation. E) Ova. F) Corpus luteum.

33. a) Menstruation. b) Pre-ovulation. c) Ovulation. d) Secretion. e) Pre-menstruation.

34.

Stage	Duration	Major occurrences
Menstruation	Day 1-5 (5 days)	- Shedding of endometrium. - Uterus bleeds. - Increase in F.S.H., oestrogen and progesterone.
Pre-ovulation	Day 5-12 (8 days)	- Endometrium repairs. - Uterus lining increases. - Development of follicle. - Increase in F.S.H and oestrogen.
Ovulation	Day 14 (1 day)	Release of ovum from mature follicle.
Secretion	Day 15-20 (6 days)	- Secretion of mucus. - Development of corpus luteum. - Increase progesterone.
Pre-menstruation	Day 21-28 (8 days)	- Degeneration of the corpus luteum and endometrium. - Decrease oestrogen and progesterone.

35. No answer supplied.

36. a) Hypothalamus, FSH-rf and Oestrogen

b) Endometrium, LH-rf and LH

c) Follicle, Ovulation and Corpus Luteum

d) Progesterone

e) Hypothalamus

f) Ovum

g) Fertilised

h) Negative feedback

37. No answer supplied

Reproductive Systems Terminology Answers

1. Fertilisation. 2. Reproduction. 3. Gametes. 4. Haploid. 5. Zygote. 6. Sexual. 7. Dimorphism.

8. Spermatogenesis. 9. Menstrual. 10. Ova. 11. Scrotum. 12. Seminiferous. 13. Fimbriae. 14. Bladder.

15. Ovary. 16. Oogenesis. 17. Uterus. 18. Interstitial. 19. Epididymis. 20. Vas Deferens. 21. Seminal.

22. Prostate. 23. Bulbo-urethral. 24. Semen. 25. Foreskin. 26. Vagina. 27. Cervix. 28. Endometrium.

29. Clitoris. 30. Labia. 31. Diploid. 32. Ovarian. 33. Puberty. 34. Menarche. 35. Menopause. 36. Ovulation.

37. Menstruation. 38. FSH. 39. LH. 40. Hypothalamus. 41. Negative. 42. Meiosis.

Differentiation & Development Answers

1. a) The fertilised ovum (combination of ovum and sperm) from which a new individual develops. b) A solid mass of cells produced as a result of multiple divisions of a fertilized ovum. c) Hollow ball of cells formed during early embryonic period.

19. a) 40% increase in blood volume plus an increase in the circulation rate by an increase in heart rate and stroke volume. b) Increase in blood flow to the lungs and breathing rate to supply extra oxygen to the foetus. c) Digestive system slows down and stomach empties slower causing constipation. d) Mother must maintain a balanced diet. The average pregnant woman needs an increase in her energy intake of about 600kJ per day especially in the second half of pregnancy. Increased intake of protein iron and calcium is also necessary. e) An increase in weight is due to foetus, placenta, amniotic fluid, breast tissue, blood volume increase etc. Optimal weight gain is no more than 500g per week in second half of a pregnancy, excess weight can be difficult to lose after birth. f) Intake of drugs should be avoided during pregnancy.
20. a) "Show" dislodgement of mucus plug from the cervical canal. b) Onset of labour pains.

21.

Stage	Duration	Major events
Dilation	Av 8-9 hours	Contractions force the head into the cervix.
Expulsion	Av. 20 mins - 2 hrs	Contractions are strong and close together the baby is pushed through the birth canal.
Placental	5 mins	Contractions of the uterus push the placenta and umbilical cord through the birth canal.

22. a) Reproductive organs slowly return to their original state (uterus returns to normal size). b) Abdomen flattens. c) Discharge from slowly shrinking uterus. d) Return of menstruation. e) Onset of lactation.
23. Before birth the lungs are not functioning to supply the foetus with oxygen and remove carbon dioxide, this role is carried out by the placenta. The ductus arteriosus and foramen ovale act to reduce blood flow to the lungs. The ductus arteriosus diverts blood from the pulmonary artery to the aorta. The foramen ovale diverts blood from the right atrium to the left atrium.
24. At birth the umbilical arteries and veins are clamped stopping blood flow through them. Blood flow through ductus arteriosus decreases and it closes up. As more blood returns from the lungs to the left atrium this increases the pressure in the left atrium closing the foramen ovale. The ventricles continue to pump blood into the aorta and pulmonary artery.

25. No answer supplied

26. a) Newborns have a large body surface area to mass, this means that they can lose or gain heat much more rapidly than larger people. Care must be taken to keep babies warm in cold conditions and cool in hot conditions. b) Newborns are prone to losing water particularly in hot conditions parents must make sure fluid intake is sufficient to compensate for this. c) Newborns are prone to infections and must be kept in a clean, dry and warm environment to reduce their chance of infection. d) Normal growth and development depends on the child's postnatal diet. Breast milk is recommended and the mother has to ensure her diet is balanced, so her milk contains all of the nutrients for normal development.
27. Genetic composition will set the limits of growth and development but the environment will determine what the final characteristics will be. For example identical twins with identical genetic composition can grow and develop differently due to variation in environmental factors eg diet and exercise.
28. a) Rooting reflex – If either cheek is touched the baby turns its head in the direction of the touch. This enables it to find the nipple of its mother's breast and together with the sucking reflex enables it to obtain food. b) Grasp reflex – A baby will automatically grasp an object placed in their hand. c) Walking reflex – If a baby is held with its feet touching the ground and moved forward it will perform walking movements. d) Startle reflex – When startled, a baby throws out its arms and legs, then pulls them back with the fingers covered as if to catch hold of something. e) Resting position – When resting on its back, a newborn baby lies with its head to one side, with the arm and leg of that side extended and the opposite arm and leg bent. f) Stepping reflex – If a baby is held with the front of one leg in contact with an object, it will raise its other leg and as if to step onto the object.

29. A – Lymphoid. B – Nervous. C – General. D – Reproductive.

Environmental Factors and Reproductive Technologies Answers

1. Smoking – causes decreases birth weight, increases risk of abortion, link between smoking and SIDS Alcohol – Can increase risk of foetal alcohol syndrome – symptoms can include lower birth weight, slow growth rate before + after birth, small head, mental retardations, facial deformities, defects of organs eg heart and limbs. A child may also suffer from behavioural problems eg poor attention, hyperactivity.
2. Teratogen causes physical defects in a developing embryo eg medicinal drugs, antibiotics, marijuana, thalidomide, some hormones, LSD, anti-tumour drugs, anticoagulants, anti convulsives.

3.

Assisted reproductive technology	Description	Advantages	Disadvantages
In-vitro fertilisation (IVF)	Fertilisation of an ovum and sperm in a glass dish in the laboratory.	Original technique assisting fertilisation in humans.	Does not address the specific fertility problem. Therefore decreased success rate.
Gamete intrafallopian transfer (GIFT)	Ova and sperm mixed in laboratory, in hope that they fertilise, then injected into the woman's uterine tube.	Couples own sperm and egg create embryo.	Only 20-30% success rate. Risk of multiple births.
Intracytoplasmic sperm injection (ICSI)	A single ovum is injected with a sperm to create an embryo which is then transplanted into the women's uterus.	Used for couples with low sperm count, or poor quality sperm.	Increased risk of birth defects.
Donor ovum	When a woman is implanted with another woman's ovum that has been mixed with he partners sperm.	Used when ovum of poor quality.	Has genes of male partner only.
Donor embryo	A donated embryo is implanted using neither partners gametes.	When both the sperm and ovum are of poor quality.	Has genes of neither parents.
Surrogacy	A woman bears a child for another couple, either using the mans sperm and her ovum, or the couples own sperm and ovum.	Couples sperm or sperm and ova can be used.	Some surrogate mothers decide they do not want to give up the child after giving birth. Causing emotional and legal problems.

4. A chemical agent or physical device/intervention used to prevent conception.

5. a) Ultrasound scans use high frequency soundwaves that 'bounce' off the foetus and body tissues forming an image. The image is in 2D and requires an experienced medical practitioner to be able to detect physical abnormalities in a developing foetus eg spina bifida, this process can also be used to determine the sex of the child. b) Foetal monitoring involves the use of devices such as ultrasound and electrocardiography to monitor a foetus heart rate. Analysis of the recording may detect oxygen deficiencies in a foetus allowing urgent procedures to be taken to avoid brain damage or death. Typically this is done before and during labour/birth to keep track of the foetus condition during this stressful period. c) In some cases due to high blood pressure or break down of the placenta it is necessary to bring forward a birth. In these cases the expectant mother is given the hormone oxytocin either intravenously or by inhalation. Oxytocin is the hormone that causes the uterus to contract, starting the birth process.

Combined pill	Contains oestrogen and progesterone, prevents release or mature egg.	Reduces incidents of ovarian and uterine cancer. Reliable.	No protection against STTs side effects possible.	Prescription needed, must be taken daily.	Excellent
Mini pill	Contains progesterone only. Making vaginal mucus thicker, thus preventing sperm entry.	Reliable, does not contain oestrogen.	No protection against STTs side effects possible.	Prescription needed, must be taken daily.	Excellent
Implants (implanon)	Slow release progesterone capsules implanted under skin.	Relatively cheap, effective and long lasting.	Prescription needed. No protection against STTs side effects possible.	Excellent	
Morning after pill	Two progesterone tablets that must be taken within 120 hours of intercourse, prevents either ovulation, fertilisation or implantation.	Available over counter, can be used if other methods haven't been used or ineffective.	Should not be used as a regular form of contraception. Ethical issues.		
Tubal ligation	Removal of small section of the uterine tubes, and the sealing of the ends.	Permanent	Usually irreversible, requires general anaesthetic.	Excellent	
Vasectomy	Removal of a small section of vas deferens open ends are then sealed to prevent passage of sperm.	Permanent	Usually irreversible, requires general or local anaesthetic.	Excellent	

Environmental Factors and Reproductive Technologies Terminology Answers

1. Teratogen. 2. Assisted reproductive technologies. 3. In-vitro fertilisation. 4. Gamete intrafallopian transfer.
5. Intracytoplasmic sperm injection. 6. Surrogate. 7. Contraception. 8. Rhythm method 9. Mechanical barrier.
10. Oestrogen and Progesterone. 11. Inter Uterine Device. 12. Bacteria. 13. Chancere 14. Abstinence/Condom
15. Chlamydia. 16. Human Immune Deficiency Virus. 17. Herpes.